

AL & Big Data Lab Report



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1. University of Orleans. (2025). Machine Learning Lab Manual Series (TPs): K-Means, DBSCAN, Hierarchical Clustering, Regression, KNN, Logistic Regression, Gradient Descent, and Perceptron. Department of Computer Science. Prepared by Prof. Prof. Frédéric Ros
2. McKinney, W. (2022). Python for Data Analysis. O'Reilly Media.

Lab 1 - K-Means Clustering

Objective

This experiment aims to explore how the K-Means clustering algorithm groups data into clusters based on similarity.

The lab focuses on:

- Understanding the K-Means process.
- Applying it to a simple 2D dataset.
- Finding the best number of clusters using the Elbow method.
- Testing it on a real-world dataset (Iris).
- Observing its strengths and weaknesses.

Theory

K-Means is an unsupervised learning algorithm used to discover natural groupings (clusters) in data. It partitions the data into K clusters such that points within a cluster are more similar to each other than to those in other clusters.

The algorithm follows these main steps:

1. **Start:** Choose K random points as initial centroids.
2. **Assign:** Allocate each data point to the nearest centroid.
3. **Update:** Recalculate centroids as the average of all points in each cluster.
4. **Repeat:** Continue until cluster centers no longer move significantly.

The algorithm minimizes the **sum of squared distances** between each point and its assigned centroid (called *inertia*).

Code

The K-Means clustering code used in this experiment is available at the following GitHub repository: [K-Means_Code](#)

Implementation on Synthetic Data

To visualize how K-Means behaves, a simple 2D dataset with three clusters was created using `make_blobs()`.

The model successfully identified three distinct clusters in the dataset. The centroids, represented by black “X” symbols, indicate the center of each cluster. Each color in the plot corresponds to a separate cluster, clearly showing separation between the data groups and confirming that the K-Means algorithm effectively grouped similar data points together.

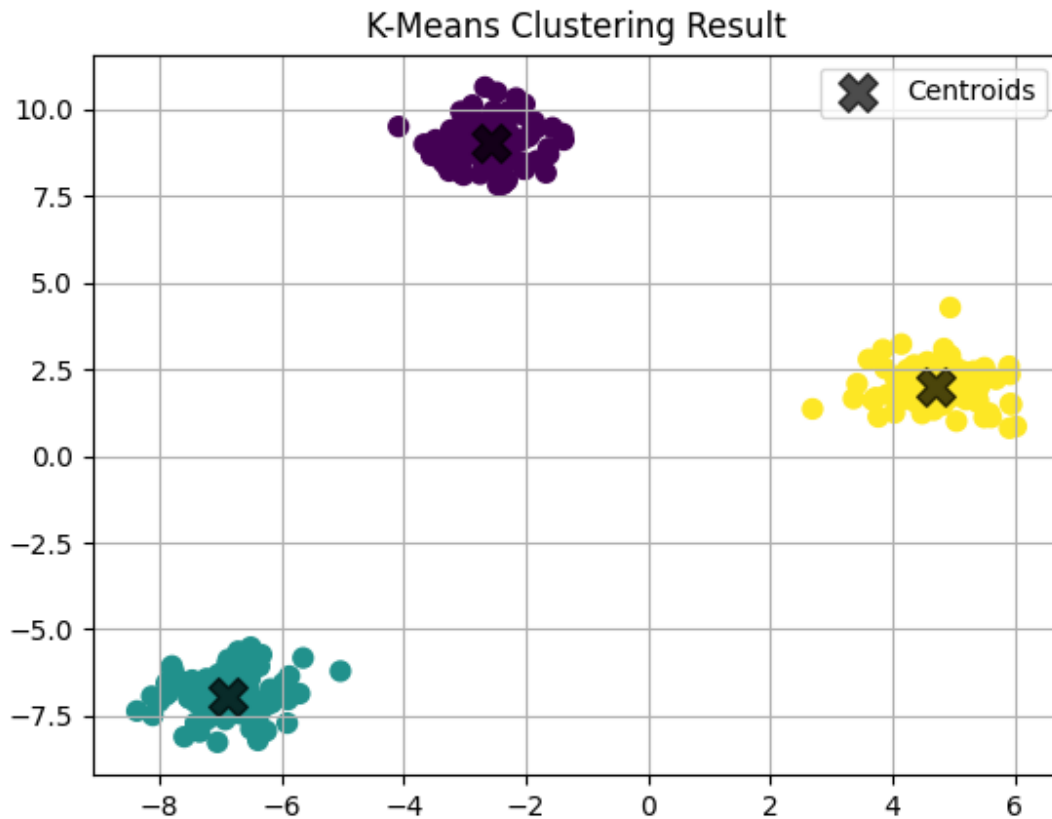


Figure 1. *K-Means clustering result on a 2D synthetic dataset.*

Determining the Optimal Number of Clusters

The Elbow Method was used to determine the most appropriate value for K. It measures how the total inertia (within-cluster sum of squares) changes as K increases.

As K grows, inertia naturally decreases because clusters become smaller and data points move closer to their respective centroids. However, beyond a certain point, the improvement becomes minimal hence additional clusters start capturing random noise rather than meaningful structure. Moreover, using too many clusters can lead to overfitting, where the model fits the data too precisely and loses generalization ability.

Therefore, the optimal value of K is chosen at the “elbow point” where adding more clusters no longer results in a significant decrease in inertia. This choice balances accuracy with simplicity, preventing the model from overfitting.

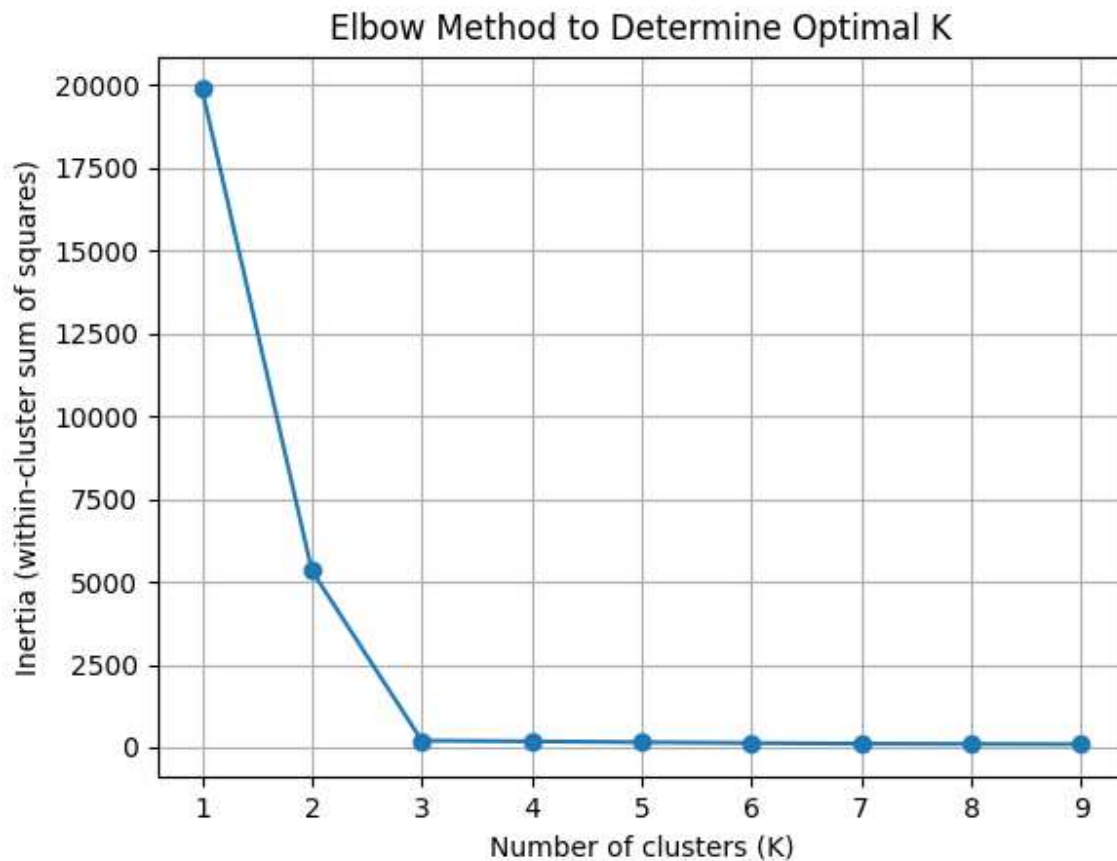


Figure 2. *Elbow Method used to determine the optimal number of clusters.*

Applying K-Means to the Iris Dataset

The algorithm was tested on the Iris flower dataset, which contains 150 samples and four numerical features.

Before clustering, the data was standardized using `StandardScaler()` to ensure that all features contributed equally to the distance calculations. This transformation scales each feature to have a mean of 0 and a standard deviation of 1, making the variables comparable across dimensions.

Principal Component Analysis (PCA) was also applied to reduce the data from four dimensions to two. This dimensionality reduction technique preserves most of the variance in the dataset while enabling effective visualization of the clustering results in a 2D plot.

K-Means successfully identified three major groups, roughly corresponding to the three Iris species. However, some overlap occurred between the purple cluster (versicolor) and the yellow cluster (virginica), indicating that the K-Means algorithm can struggle when clusters overlap or are not linearly separable in feature space.

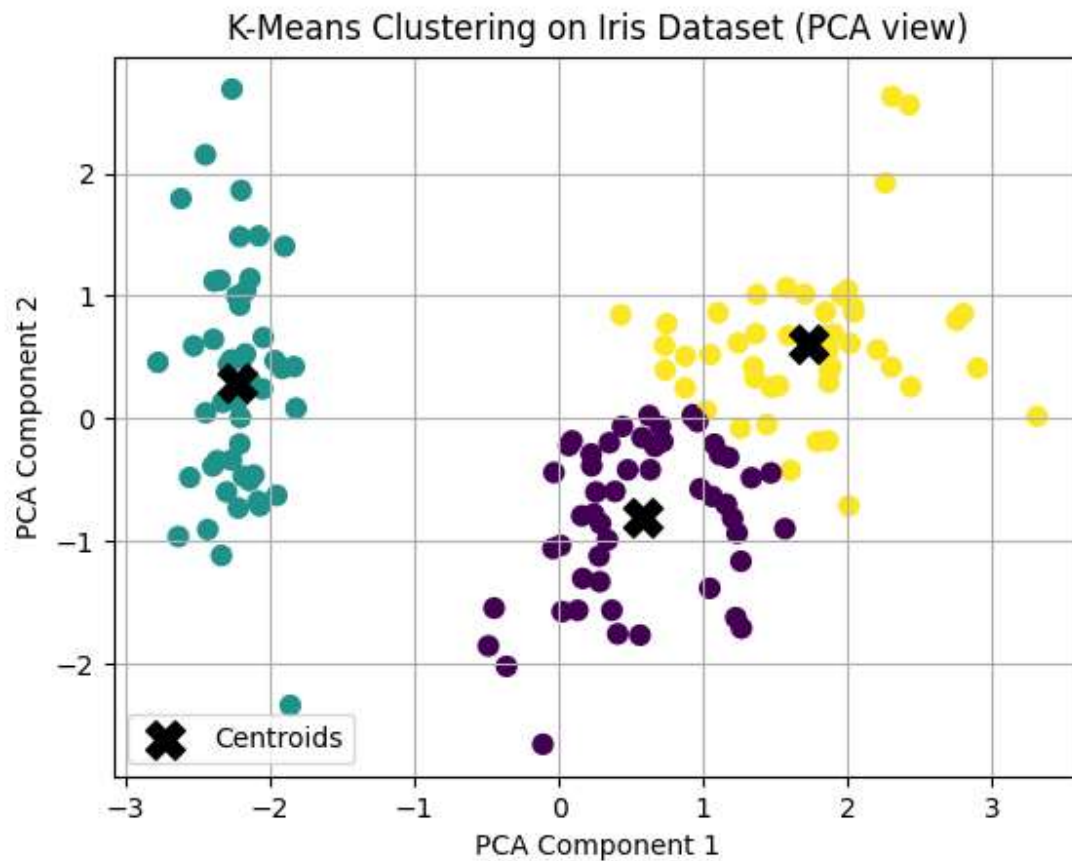


Figure 3. *K-Means clustering on the Iris dataset (2D PCA projection).*

Exploring Limitations with Non-Spherical Data

To examine how K-Means performs on irregularly shaped datasets, two synthetic datasets were generated. One was shaped like interlocking moons and another like concentric circles.

In both cases, the algorithm failed to correctly separate the curved clusters because K-Means relies on Euclidean distance to assign points to the nearest centroid. As a result, it naturally forms circular cluster boundaries. This assumption works well for spherical data but performs poorly when clusters are non-linear or non-spherical. Consequently, K-Means misclassifies several data points in regions where the clusters bend or overlap.

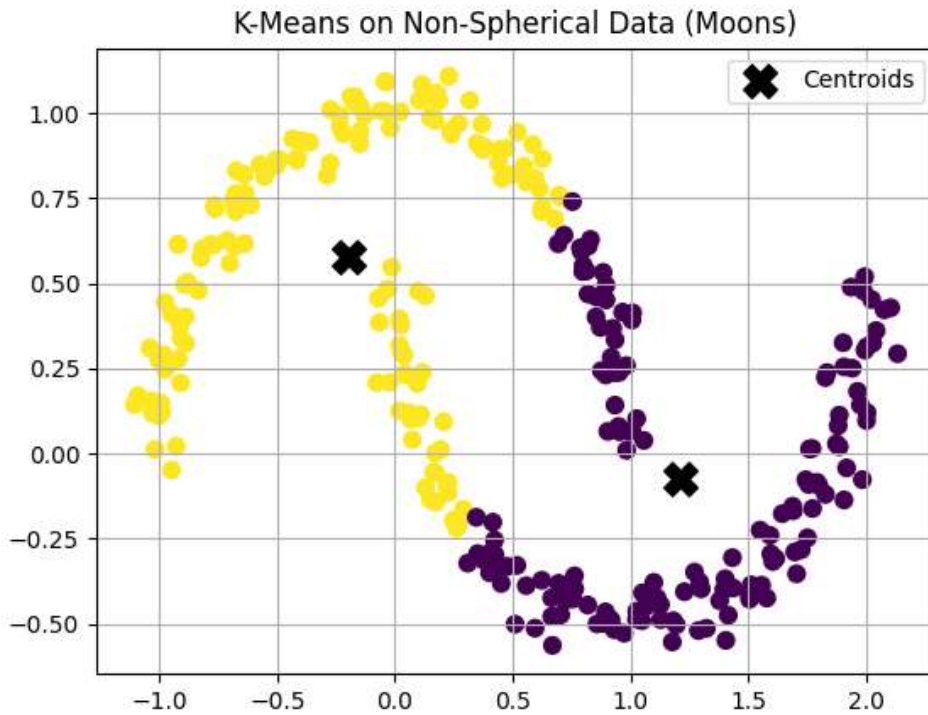


Figure 4. *K-Means clustering on non-spherical data (two interlocking moons)*

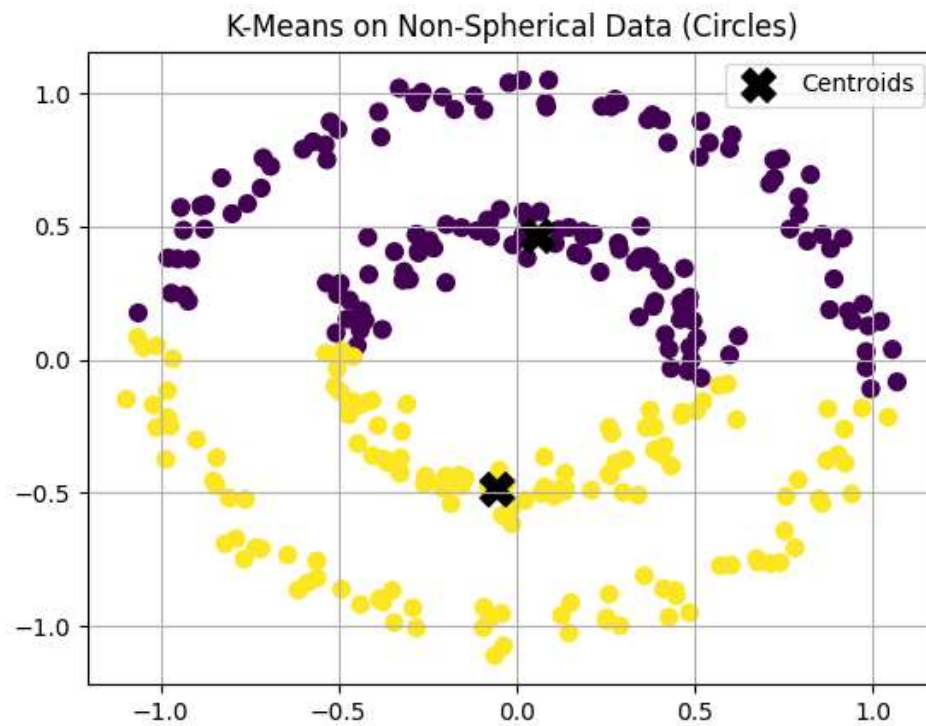


Figure 5. *K-Means clustering on non-spherical data (concentric circles)*

Cluster Evaluation (Silhouette Score)

The Silhouette Score measures how well each data point fits within its assigned cluster compared to other clusters. The score ranges from -1 to 1, where a value closer to 1 indicates that the data points are well clustered, while lower or negative values suggest poor separation or incorrect clustering. The highest silhouette value was observed at $K = 2$, indicating that the data is most cleanly divided into two distinct clusters. However, $K = 3$ still provides a reasonable structure that aligns with the known Iris species groupings.



Figure 6. Silhouette Score versus number of clusters for the Iris dataset.

Questions for Discussion

1. What are the limitations of K-Means?
K-Means requires specifying K in advance, assumes spherical and equally sized clusters, and performs poorly on irregular, overlapping, or outlier-heavy data.
2. What happens if clusters are not spherical?
K-Means forces circular boundaries, so it struggles with curved or uneven clusters, often splitting points from the same true group or grouping distant points together incorrectly.
3. Does K-Means always find the best result?
No. K-Means can get stuck in a local minimum, meaning it may find a suboptimal solution depending on where the centroids were initially placed.

4. What if centroids are initialized poorly?

Poor initialization can lead to unbalanced or incorrect clusters and may also cause slower convergence. The K-Means++ method reduces this issue by choosing well-spaced initial centroids for more stable and accurate results.

Conclusion

K-Means is one of the most widely used and intuitive algorithms for identifying structure in data. It performs well on compact, well-separated clusters but struggles with irregular, overlapping, or non-spherical shapes. Overall, it serves as an excellent starting point for clustering analysis, though more advanced algorithms such as DBSCAN or Gaussian Mixture Models (GMMs) are better suited for complex or non-linear datasets.

K-Means is most commonly applied in customer segmentation, image compression, document grouping, and market analysis, where cluster boundaries are approximately convex and the number of groups can be estimated beforehand.

Lab 2 - DBSCAN Clustering

Objective

This experiment aims to explore how the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm discovers clusters by local density and identifies outliers. The lab focuses on:

- Understanding DBSCAN's core ideas and parameters (**eps**, **min_samples**) and the roles of core, border, and noise points.
- Applying DBSCAN to a non-spherical 2D dataset (two moons) and examining parameter sensitivity.
- Comparing DBSCAN with K-Means on the same data to highlight shape and outlier handling differences.
- Running DBSCAN on the Iris dataset (with PCA for visualization) and counting clusters vs. noise points.
- Noting practical limitations (single global **eps**, varying densities) and when alternatives like HDBSCAN may be preferable.

Theory

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a density-based clustering algorithm that groups together points closely packed in space, while marking isolated points as noise. It defines clusters based on two parameters:

- ϵ (eps): the neighborhood radius around a point.
- min_samples: the minimum number of points required to form a dense region.

Conceptually, the method considers a neighborhood of radius ϵ around each point in the dataset. If this neighborhood contains at least min_samples points, the point is classified as a **core point**. Points that lie within the ϵ -neighborhood of a core point but do not themselves meet the density requirement are referred to as **border points**. Points that are neither core nor border points are designated as **noise** or **outliers**.

Clusters are then constructed by connecting core points whose neighborhoods overlap, thereby forming continuous regions of high density. Unlike K-Means, DBSCAN does not require specifying the number of clusters and can identify clusters of arbitrary shapes, making it robust for non-spherical data and outlier detection. However, its performance depends strongly on suitable ϵ and min_samples values.

Code

The DBSCAN clustering code used in this experiment is available at the following GitHub repository: [DBSCAN_Code](#)

Implementation on Synthetic Non-Spherical Data

A two-dimensional non-spherical dataset was generated using the `make_moons()` function with 300 samples and a noise level of 0.05. This dataset was selected because it cannot be effectively separated by algorithms that assume spherical clusters, such as K-Means. The DBSCAN algorithm was applied with parameters $\epsilon = 0.2$ and `min_samples = 5` to identify the two crescent-shaped clusters.

These parameter values were selected empirically through visual inspection of the distance distribution, as they provided a balance between accurately detecting the two main clusters and minimizing the number of points classified as noise.

A smaller ϵ value tends to fragment the data into numerous small clusters, often increasing the number of noise points and resulting in underfitting, where the model is too rigid to capture the full cluster structure. Conversely, an excessively large ϵ may merge distinct regions into a single cluster, leading to overfitting, where the model becomes too general and fails to preserve meaningful boundaries. Similarly, a low `min_samples` value allows small or spurious clusters to form, while a high `min_samples` value enforces stricter density requirements that may exclude legitimate but less dense clusters.

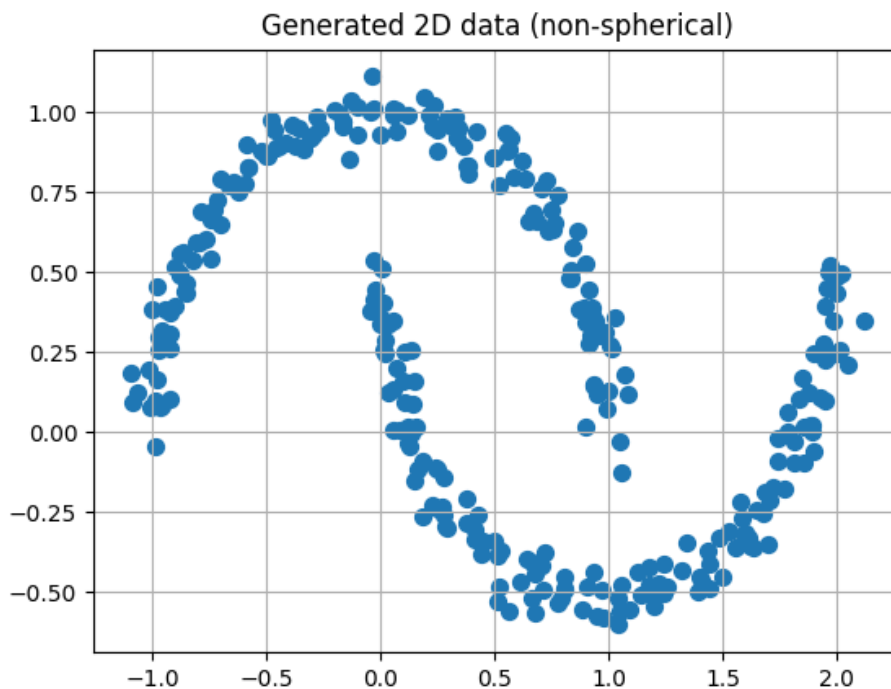


Figure 7. Generated two-dimensional non-spherical dataset (two-moons) created using the `make_moons()` function.

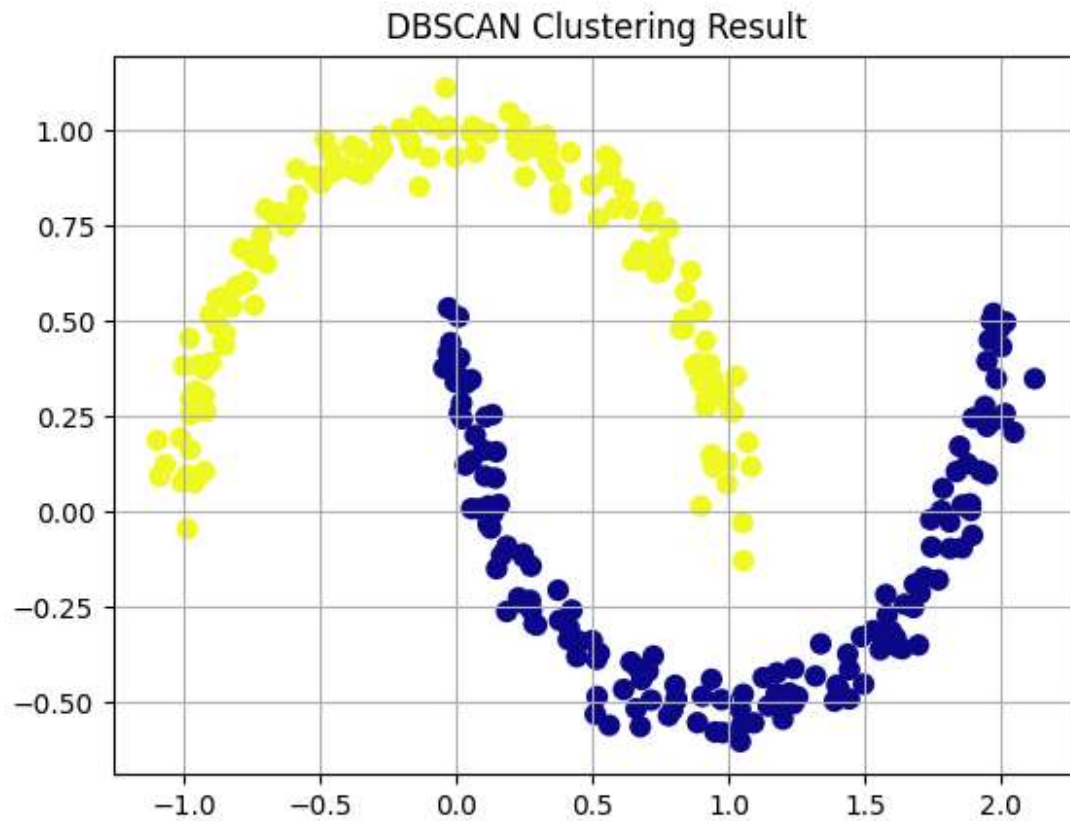


Figure 8. DBSCAN clustering result on the two-moons dataset using parameters $\epsilon = 0.2$ and `min_samples = 5`.

Parameter Sensitivity

To examine the influence of DBSCAN's parameters, experiments were conducted by varying ϵ and `min_samples` independently.

As ϵ increased, the neighborhood radius expanded, causing the algorithm to transition from detecting multiple small, fragmented clusters (when ϵ was small) to merging the two crescent-shaped structures into a single cluster (when ϵ was large). Smaller ϵ values led to excessive fragmentation and noise (underfitting), while larger ϵ values caused overgeneralization and cluster merging (overfitting).

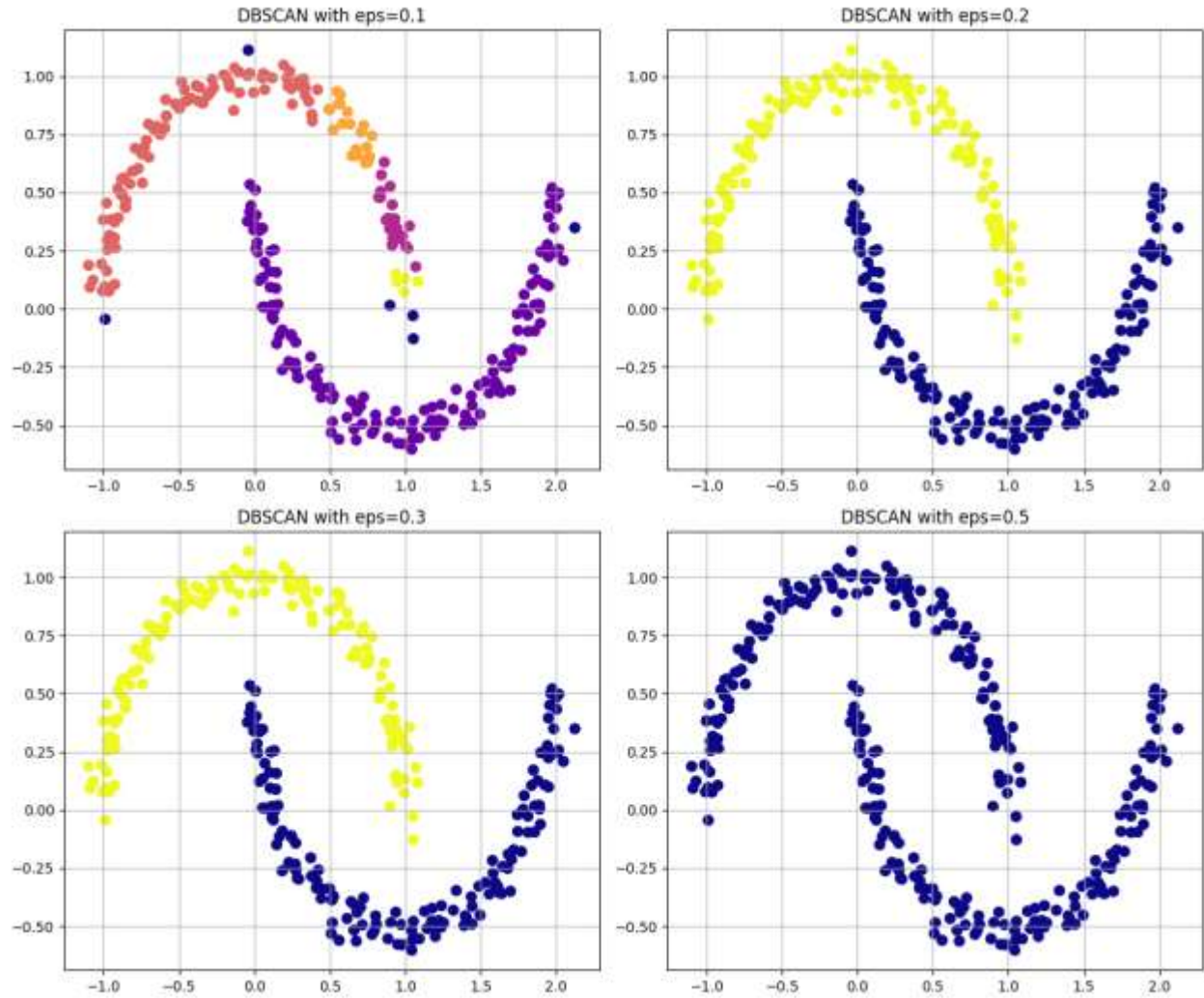


Figure 9. DBSCAN results with varying ϵ values (0.1, 0.2, 0.3, 0.5) and fixed $\text{min_samples} = 5$.

Similarly, increasing min_samples made the density criterion progressively stricter, reducing the number of core points and increasing noise, especially at higher values. Smaller min_samples values (e.g., 3–5) produced stable and coherent clusters, while very large values (e.g., 20) led to overly conservative clustering that excluded many valid points at the edges of each crescent.

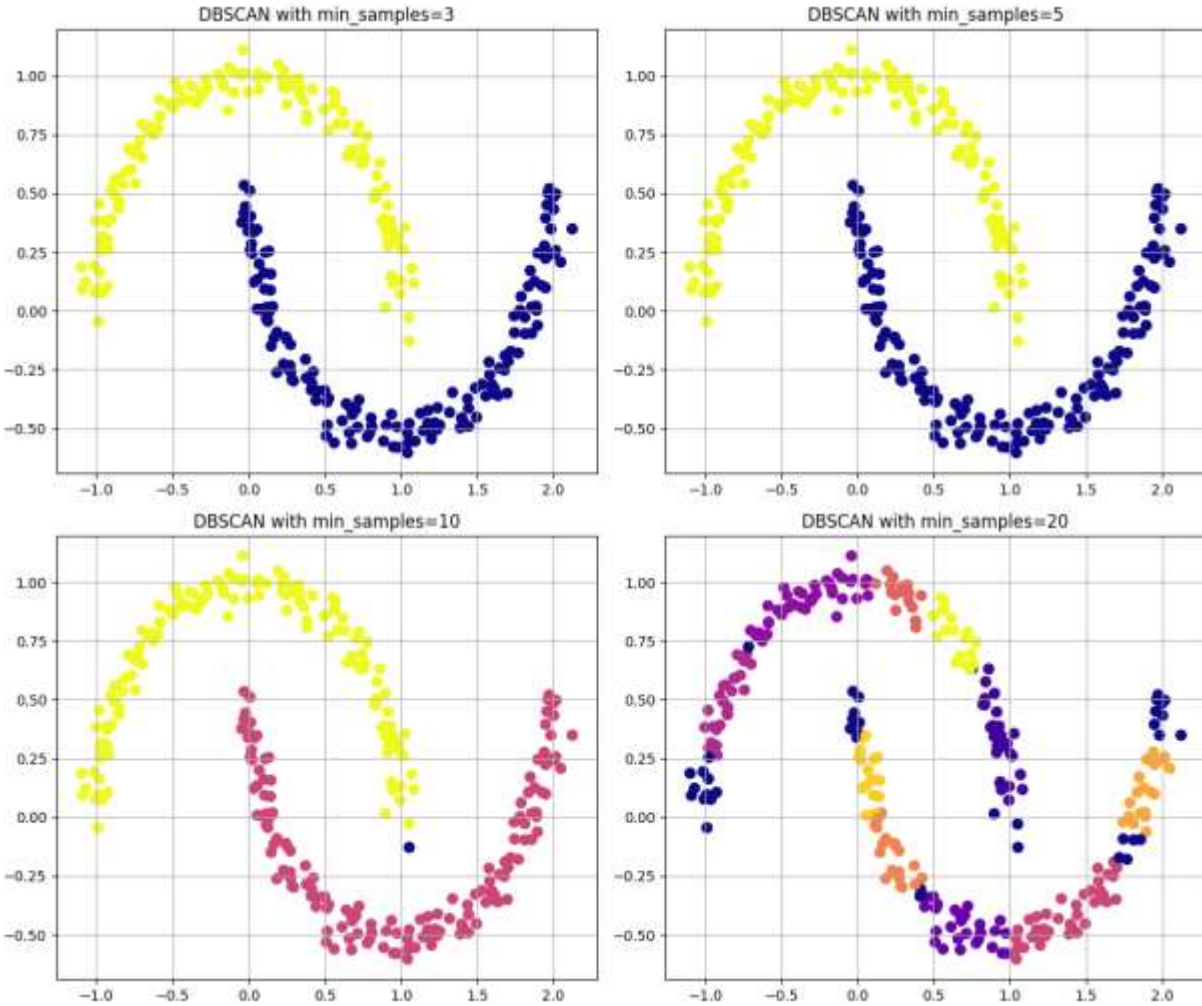


Figure 10. DBSCAN results with varying `min_samples` values (3, 5, 10, 20) and fixed $\epsilon = 0.2$.

An intermediate configuration ($\epsilon = 0.2$, `min_samples` = 5) yielded the most coherent and balanced clustering.

Comparing to K-Means

The same non-spherical two-moons dataset was also clustered using K-Means with $k = 2$. K-Means divided the data into two regions using straight, centroid-based boundaries that do not follow the curved structure of the dataset. As a result, several points near the junction of the two crescents were misclassified due to the algorithm's assumption of spherical cluster geometry and its reliance on Euclidean distance to fixed centroids.

DBSCAN, in contrast, accurately identified the two curved clusters without requiring the number of clusters in advance. Its density-based approach captured the true manifold of the data and correctly separated the clusters while labeling a few boundary points as noise. This result demonstrates DBSCAN's ability to model irregular and non-convex shapes that K-Means cannot effectively represent.

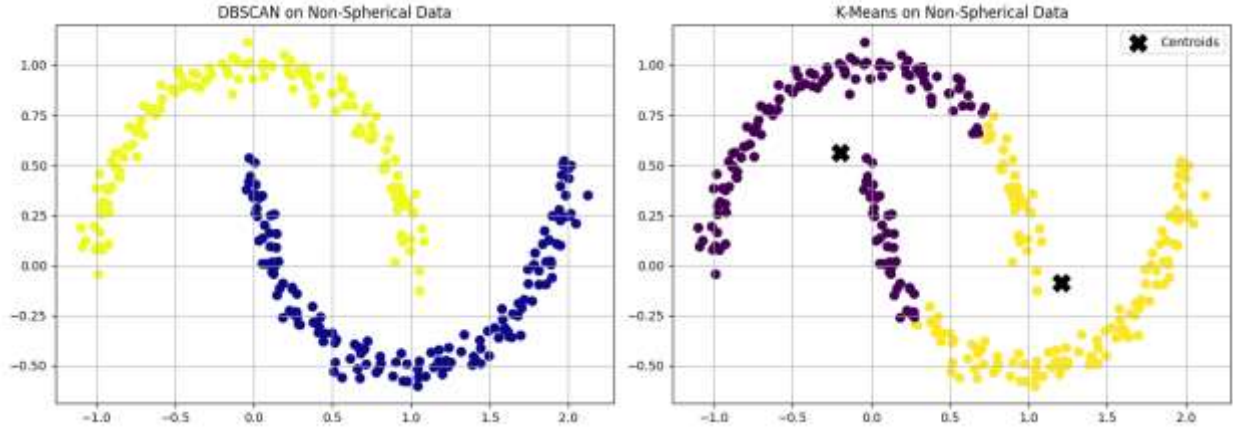


Figure 11. Comparison of clustering performance on the non-spherical two-moons dataset between DBSCAN ($\epsilon = 0.2$, $\text{min_samples} = 5$) on the left and K-Means ($k = 2$) on the right.

DBSCAN on the Iris Dataset (Real-World Multidimensional Data)

The Iris dataset was used to evaluate DBSCAN on real-world, multi-dimensional data. The feature variables were standardized using StandardScaler to ensure equal contribution of all attributes to the distance metric. Dimensionality reduction was then performed using Principal Component Analysis (PCA) to project the four-dimensional data into two principal components for visualization purposes only.

DBSCAN was applied to the standardized data using parameters $\epsilon = 0.6$ and $\text{min_samples} = 5$. The algorithm identified several compact regions of high density corresponding to potential clusters while labeling certain ambiguous samples as noise. The resulting two-dimensional PCA projection revealed visually separable groups that broadly corresponded to the Iris species, although some overlap occurred between Iris versicolor and Iris virginica, which are known to have similar feature distributions.

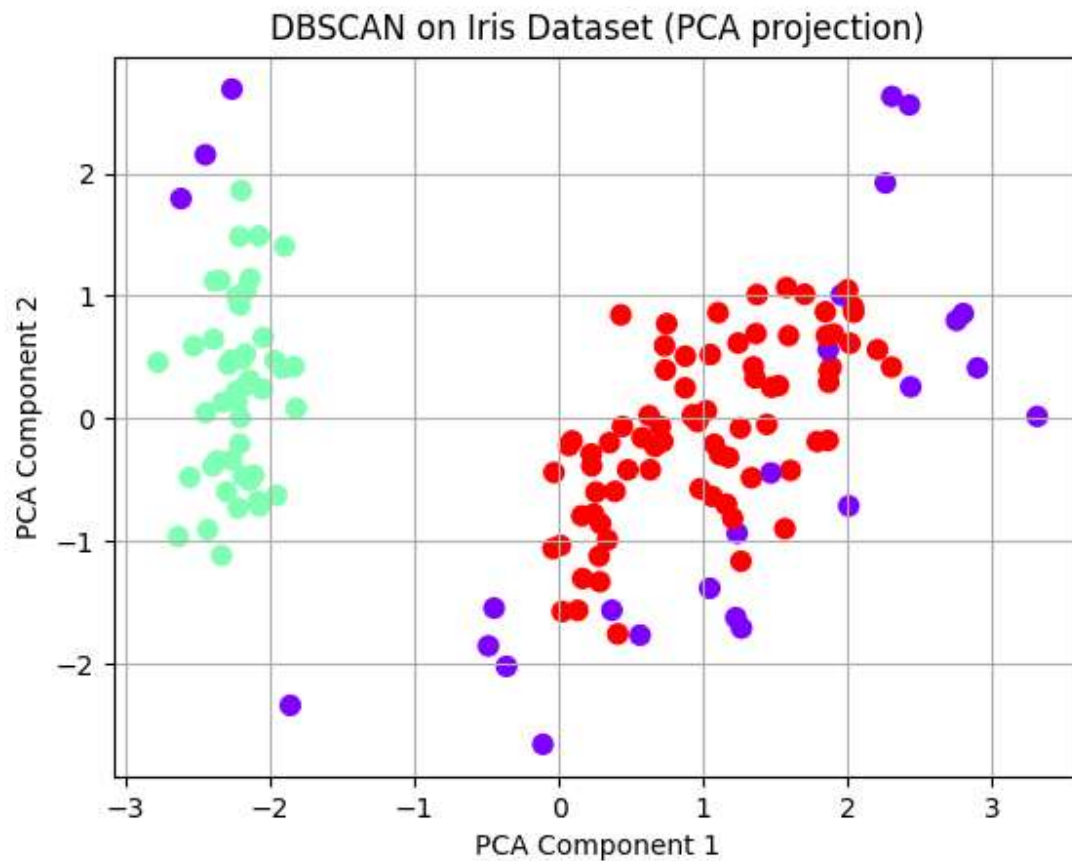


Figure 12: DBSCAN clustering of the Iris dataset visualized after PCA projection ($\epsilon = 0.6$, $\text{min_samples} = 5$).

Data with Varying Density

To examine DBSCAN's behaviour when cluster densities differ, a synthetic dataset was generated using the `make_blobs()` function with cluster standard deviations of $[0.5, 1.5, 0.3]$, creating three groups of unequal density. The algorithm was executed with parameters $\epsilon = 0.8$ and $\text{min_samples} = 5$.

The resulting plot shows that DBSCAN correctly identifies the two denser clusters but partially merges or misclassifies points in the sparser, more diffuse cluster. This occurs because DBSCAN applies a single global density threshold (ϵ), which cannot simultaneously accommodate both dense and sparse regions. Consequently, points belonging to low-density clusters are either labeled as noise or absorbed into neighboring clusters.

This experiment highlights DBSCAN's sensitivity to varying density and motivates the use of adaptive approaches such as HDBSCAN, which can handle non-uniform density distributions by adjusting neighborhood thresholds locally.

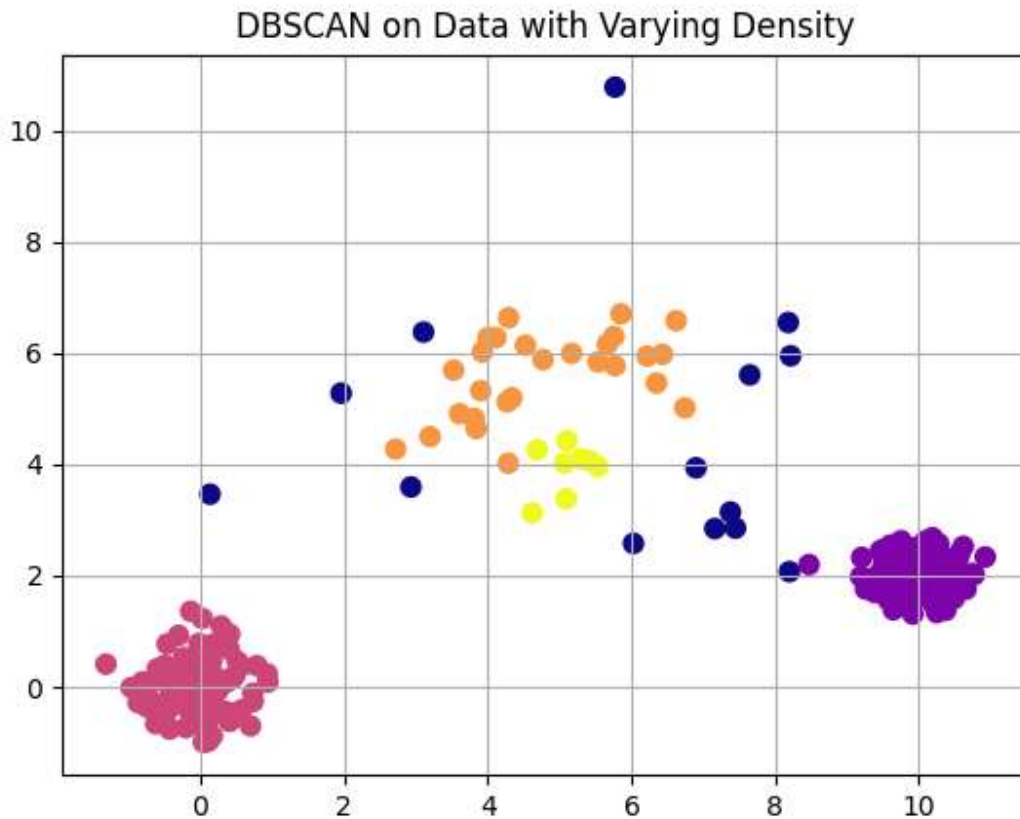


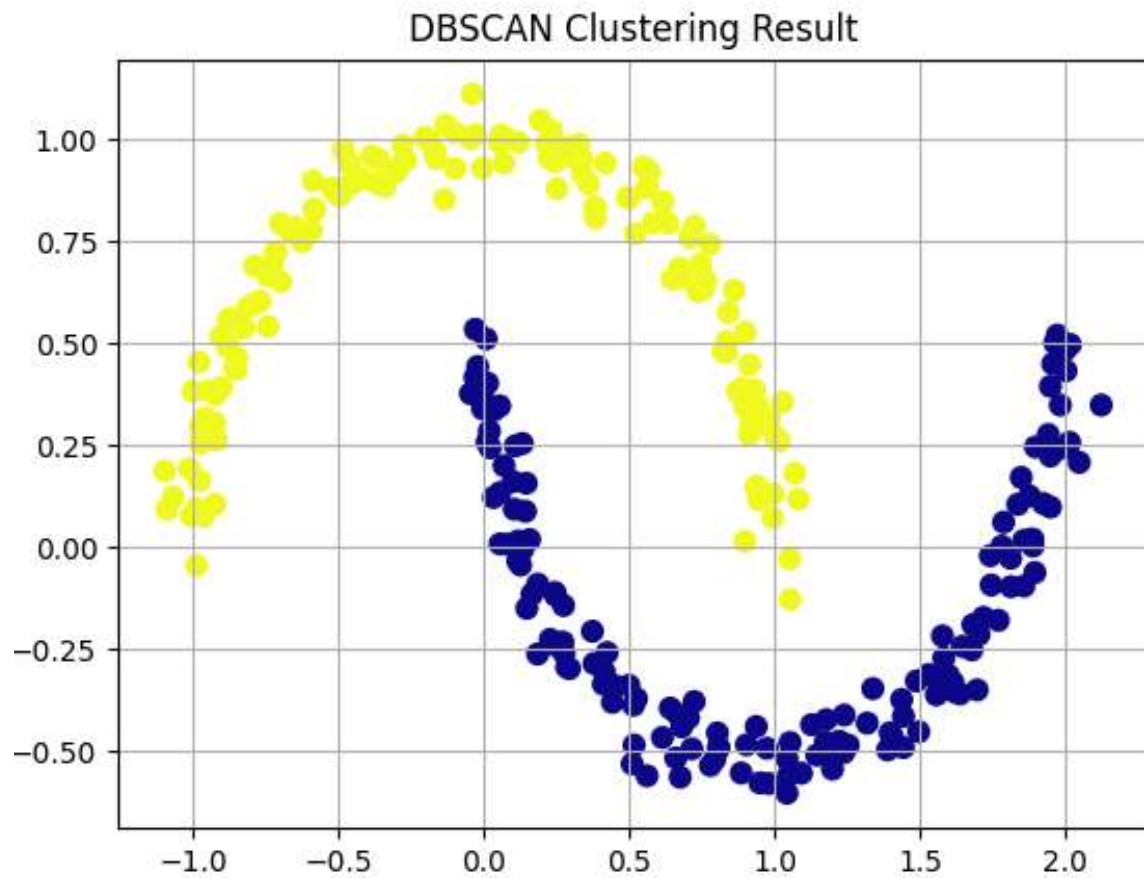
Figure 13: DBSCAN clustering of the Iris dataset visualized after PCA projection ($\epsilon = 0.6$, $\text{min_samples} = 5$).

Using Silhouette Score and Adjusted Rand Index to Evaluate Clustering Quality

To quantitatively assess DBSCAN's performance, two standard clustering evaluation metrics were computed on the two-moons dataset: the Silhouette Score and the Adjusted Rand Index (ARI).

The Silhouette Score of 0.328 indicates moderate cluster separation. Points within each cluster are generally closer to one another than to points in other clusters. However, some points near the edges of the crescent-shaped regions lie close in Euclidean distance to points in the neighboring cluster. Because the Silhouette metric evaluates separation based purely on straight-line distances, it penalizes these boundary points even though they are correctly assigned. This results in a moderate score, which is expected for non-spherical data such as the two-moons structure.

The Adjusted Rand Index (ARI) achieved a perfect score of 1.000, confirming that DBSCAN's cluster assignments exactly matched the true underlying structure of the dataset. This result demonstrates DBSCAN's strength in identifying complex, non-spherical cluster shapes that are difficult for centroid-based algorithms such as K-Means to represent accurately.



```
Run 2. dbscan x
Number of clusters <Moons Dataset>: 2
Number of noise points <Moons Dataset>: 0
Silhouette Score: 0.328
Adjusted Rand Index: 1.000
Process finished with exit code 0
```

Figure 14: Quantitative evaluation of DBSCAN performance on the two-moons dataset ($\epsilon = 0.2$, $\text{min_samples} = 5$).

HDBSCAN on Varying-Density Data

To address DBSCAN's limitation in handling clusters of unequal density, the same synthetic dataset with varying cluster standard deviations $[0.5, 1.5, 0.3]$ was analyzed using HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise). Unlike DBSCAN, HDBSCAN does not rely on a single global ϵ value; instead, it builds a hierarchy of density levels and extracts stable clusters based on their persistence across scales.

With parameters `min_cluster_size = 15` and `min_samples = 5`, HDBSCAN successfully detected three main clusters and a small number of noise points. Compared to DBSCAN, which struggled to separate the sparser middle group, HDBSCAN adaptively adjusted to local density variations, resulting in more accurate cluster boundaries and reduced misclassification of low-density points as noise.

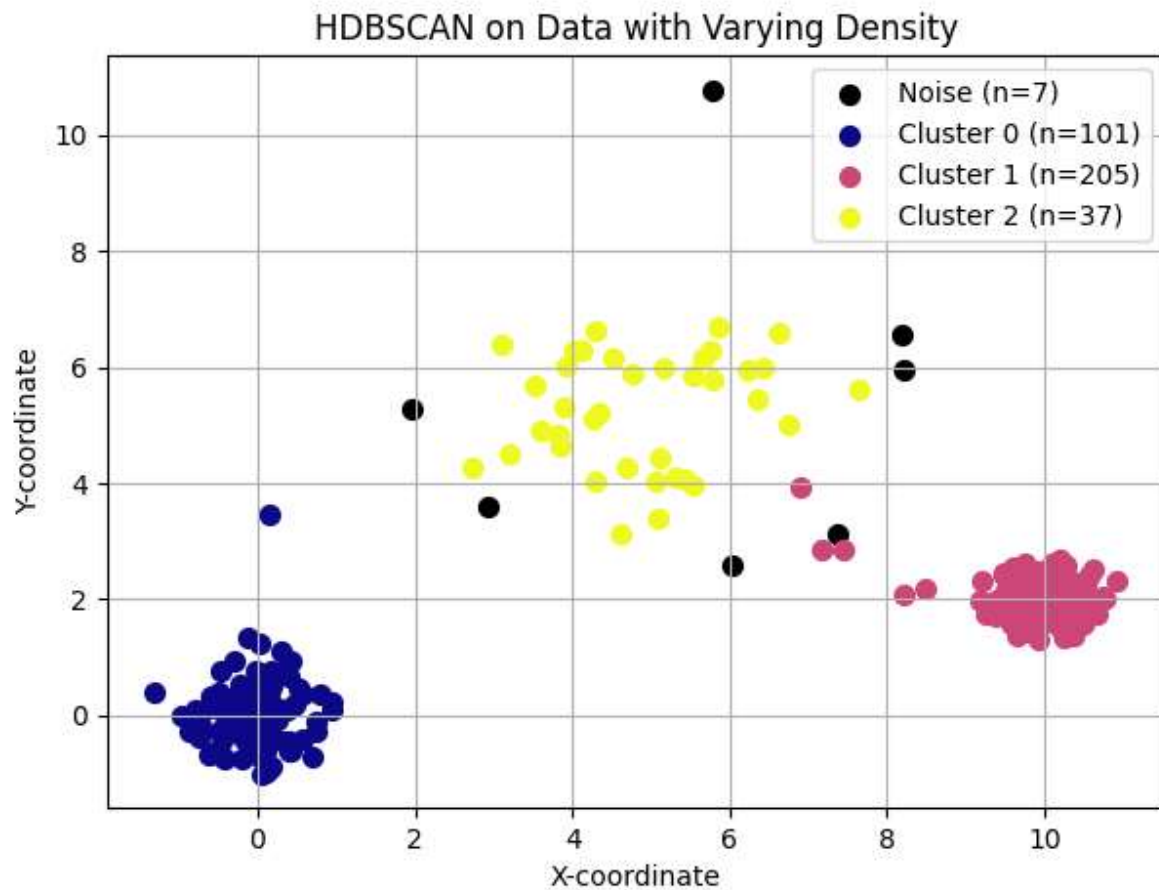


Figure 15: HDBSCAN clustering on data with varying density (`min_cluster_size = 15`, `min_samples = 5`).

Questions for Discussion

1. What are the advantages of DBSCAN over K-Means?
DBSCAN does not require the number of clusters to be specified in advance and can discover clusters of arbitrary shape, such as non-spherical or irregular structures that K-Means cannot represent. It is also able to detect outliers and label them as noise.
2. What are the limitations of DBSCAN?
DBSCAN is sensitive to its parameters ϵ (epsilon) and `min_samples`. Small changes in these values can significantly affect the resulting clusters. It also struggles with datasets containing clusters of varying density, since a single global ϵ cannot accommodate both dense and sparse regions simultaneously. In addition, DBSCAN's performance can degrade on very large or high-dimensional datasets because distance measures become less meaningful in such spaces.
3. Can DBSCAN identify noise in the Iris dataset?
Yes. DBSCAN labels points that do not belong to any sufficiently dense region as noise (label = -1).
4. Would DBSCAN work well in high-dimensional datasets?
Not very well. In high-dimensional spaces, the notion of "distance" becomes less informative due to the curse of dimensionality, making it difficult to define meaningful density thresholds. As a result, DBSCAN may either classify most points as noise or merge clusters incorrectly. Dimensionality reduction techniques such as PCA or t-SNE are often applied before using DBSCAN on high-dimensional data.

Conclusion

Through experiments on both synthetic and real-world datasets, DBSCAN was shown to effectively identify clusters of arbitrary shape without requiring the number of clusters to be specified in advance. Its ability to label low-density points as noise further distinguishes it from centroid-based algorithms such as K-Means, which assume spherical clusters and assign all data points to a group.

However, DBSCAN's performance was found to be highly dependent on the choice of its parameters, ϵ and `min_samples`, and it struggled when clusters exhibited varying densities. In contrast, HDBSCAN addressed this limitation by adaptively determining density thresholds, yielding more stable results and better separation of clusters with differing densities.

DBSCAN is most commonly applied in scenarios where data exhibit irregular shapes, variable densities, or contain noise that should not be forced into clusters. It is widely used in spatial data analysis, such as identifying geographic regions of interest or detecting hotspots in GPS data, as well as in anomaly and outlier detection in network security, manufacturing, and sensor data. DBSCAN is also applied in image processing, genomics, and astronomical data analysis, where natural groupings are complex and not easily captured by algorithms assuming spherical clusters or uniform densities.

Lab 3 - Hierarchical Clustering

Objective

This experiment aims to explore how hierarchical clustering groups data by progressively merging similar observations to form a tree-structured representation of clusters. The lab focuses on:

- Understanding the principles of agglomerative hierarchical clustering (the bottom-up approach), which is the most commonly used method.
- Visualizing how clusters are formed through a dendrogram, which reveals the order and distance at which points or groups are merged.
- Examining the effect of different linkage criteria (single, complete, average, and Ward) and how they influence cluster shapes and merging behavior.
- Applying hierarchical clustering to both synthetic 2D data and a real-world dataset (Iris), to perform dimensionality reduction using PCA for visualization.
- Investigating methods for extracting clusters from a dendrogram and comparing hierarchical clustering with K-Means and DBSCAN on the same dataset.

Theory

Hierarchical clustering is an unsupervised learning technique that groups data based on similarity while preserving the structure of how clusters are formed. Unlike partitioning algorithms such as K-Means, it does not require specifying the number of clusters in advance. Instead, hierarchical clustering organizes data into a multi-level hierarchy, often visualized as a tree-like structure called a dendrogram.

There are two main approaches:

1. Agglomerative clustering (bottom-up):

This method starts by treating each data point as its own cluster. At each step, the two closest clusters are merged based on a chosen distance metric and linkage criterion. This process continues until all points form a single cluster. Agglomerative clustering is the most widely used approach and the one explored in this lab.

2. Divisive clustering (top-down):

This approach begins with all data points in a single cluster and iteratively splits clusters into smaller ones. Although conceptually opposite to the agglomerative method, divisive clustering is less common due to its higher computational cost.

To decide which clusters to merge (or split), hierarchical clustering relies on a linkage criterion, which determines how distances between clusters are calculated. The most common linkage methods include:

1. **Single linkage:** Single linkage calculates the distance between two clusters by looking at the closest pair of points across them. This approach often creates long chain-like clusters because a single close connection can link two otherwise distant groups.
2. **Complete linkage:** Complete linkage measures the distance between the farthest points in each cluster. This method produces compact and well-separated clusters, but it can sometimes split clusters that lie close to each other.
3. **Average linkage:** Average linkage uses the mean distance between all pairs of points across the two clusters. This approach provides a balanced behaviour, avoiding the chaining effect of single linkage and the strict compactness of complete linkage.
4. **Ward linkage:** Ward linkage merges clusters in a way that causes the smallest increase in total within-cluster variance. It often creates the most compact and well-structured clusters, which makes it a common default choice in many applications.

Hierarchical clustering is most useful when the goal is not only to cluster data but also to understand the relationships between clusters, visualize how they form, and explore structure at multiple granularity levels. However, it can be computationally expensive for large datasets and may be sensitive to noise depending on the linkage method used.

Code

The Hierarchical Clustering Code used in this experiment is available at the following GitHub repository: [Hierarchial Clustering Code](#)

Synthetic Data and Clustering

To begin the experiment, a simple two-dimensional dataset was generated using the `make_blobs` function. The dataset contains 200 samples grouped around three well-separated cluster centers, each with a moderate standard deviation. This setup provides a clear structure that allows hierarchical clustering to reveal how points gradually merge into larger groups.

The scatter plot of the generated data shows three compact and visually distinct clusters. Each point represents an individual sample in the dataset. Because the clusters are well defined and do not overlap, this synthetic example offers an ideal starting point for observing how the hierarchical algorithm progressively builds the clustering hierarchy during the merging process.

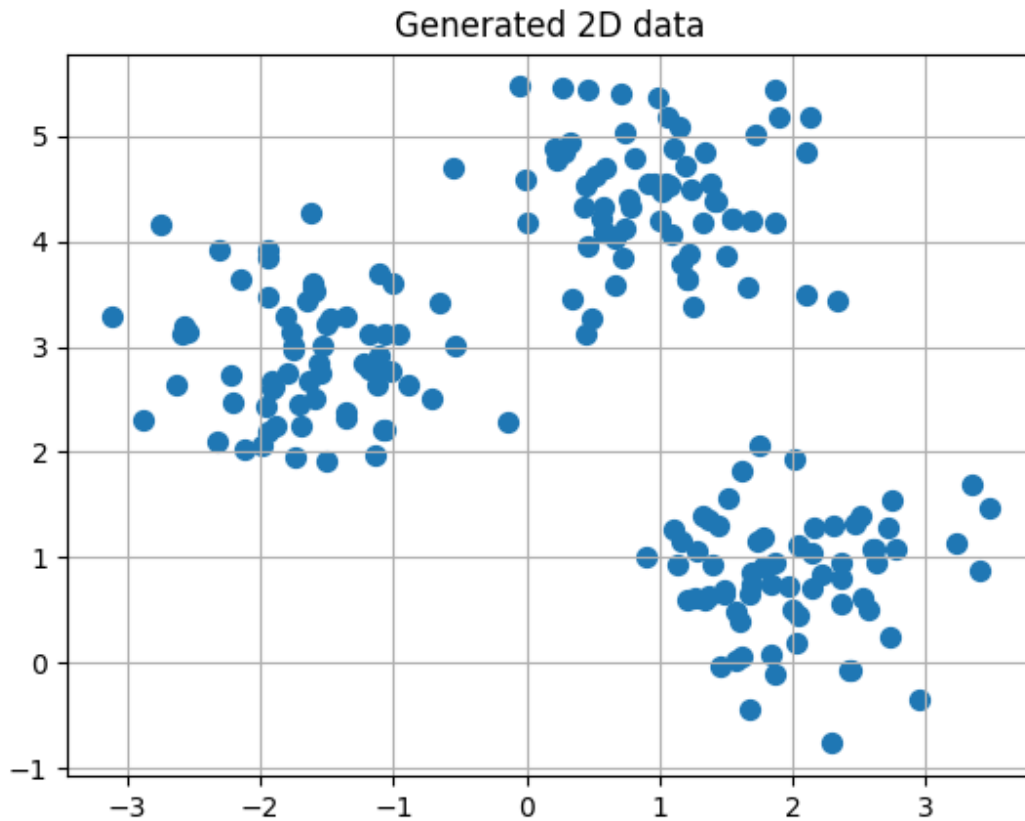


Figure 16. *Generated 2D synthetic dataset.*

Agglomerative Clustering with Dendrogram

After generating the synthetic dataset, the next step involves applying agglomerative hierarchical clustering and visualizing how the clusters merge over time. The algorithm begins with each data point treated as an individual cluster. At each iteration, it identifies the two clusters that lie closest to each other according to the chosen linkage method (Ward's method) and merges them. This repeated merging gradually builds the full hierarchical structure.

The dendrogram provides a visual summary of this merging process. Each horizontal line in the diagram represents a merge between two clusters, and the height of the line reflects the distance at which the merge occurs. Lower merges indicate pairs of points or clusters that lie close together, while higher merges correspond to groups that are more dissimilar. By examining the overall structure of the dendrogram, it becomes possible to see how the three main clusters form and at what distances they join.

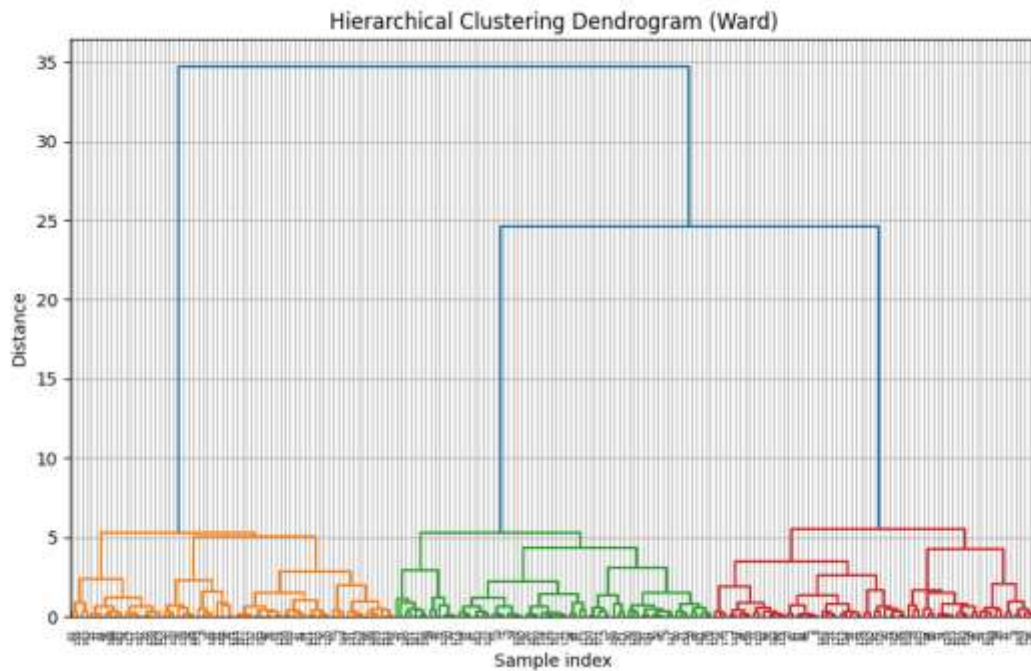


Figure 17. *Hierarchical clustering dendrogram (Ward linkage).*

Cluster Extraction

Once the hierarchical structure is built, the next step is to extract a specific number of clusters from the dendrogram. In this experiment, the agglomerative algorithm uses Ward linkage and forms three clusters, which matches the natural grouping in the synthetic dataset. By specifying $n_clusters = 3$, the algorithm stops the merging process at the appropriate level and assigns each data point to one of the resulting clusters.

The scatter shows the final clustering outcome. Each color represents one of the three clusters identified by the algorithm. The model clearly separates the data into the same groups visible in the original dataset, and the cluster boundaries align well with the natural structure. This result confirms that hierarchical clustering can accurately recover well-defined groups and provides a transparent merging path through the dendrogram.

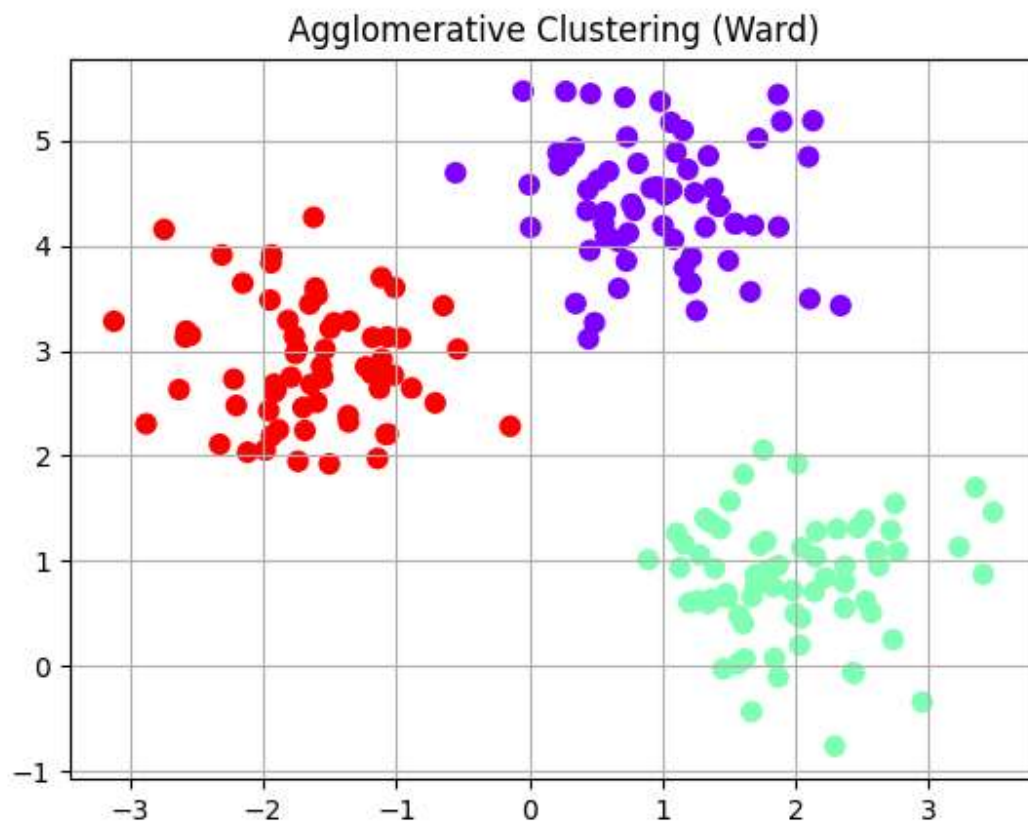


Figure 18. *Agglomerative clustering result using Ward linkage.*

Comparing Linkage Methods

Different linkage methods create noticeably different clustering results because each method uniquely calculates the distance between clusters. In this part of the lab, four linkage strategies (single, complete, average, and Ward) were applied to the same synthetic dataset from the earlier sections. Each method attempted to merge clusters based on similarity, but each one used a different definition of distance, which led to variations in cluster shapes and boundaries. These differences showed how strongly the choice of linkage influenced the structure of the final clustering result.

Figure 19 shows how the algorithm partitions the data under each linkage method. Single linkage often produces elongated, chain-like clusters because it merges groups as soon as any pair of points lies close together. Complete linkage behaves more conservatively by considering the farthest points between clusters, which results in compact and well-separated groups but can sometimes split clusters that sit close to one another. Average linkage balances these two extremes by merging clusters based on the mean distance between all point pairs across them. Ward linkage typically generates the most compact and stable clusters because it minimizes the growth of within-cluster variance at every step.

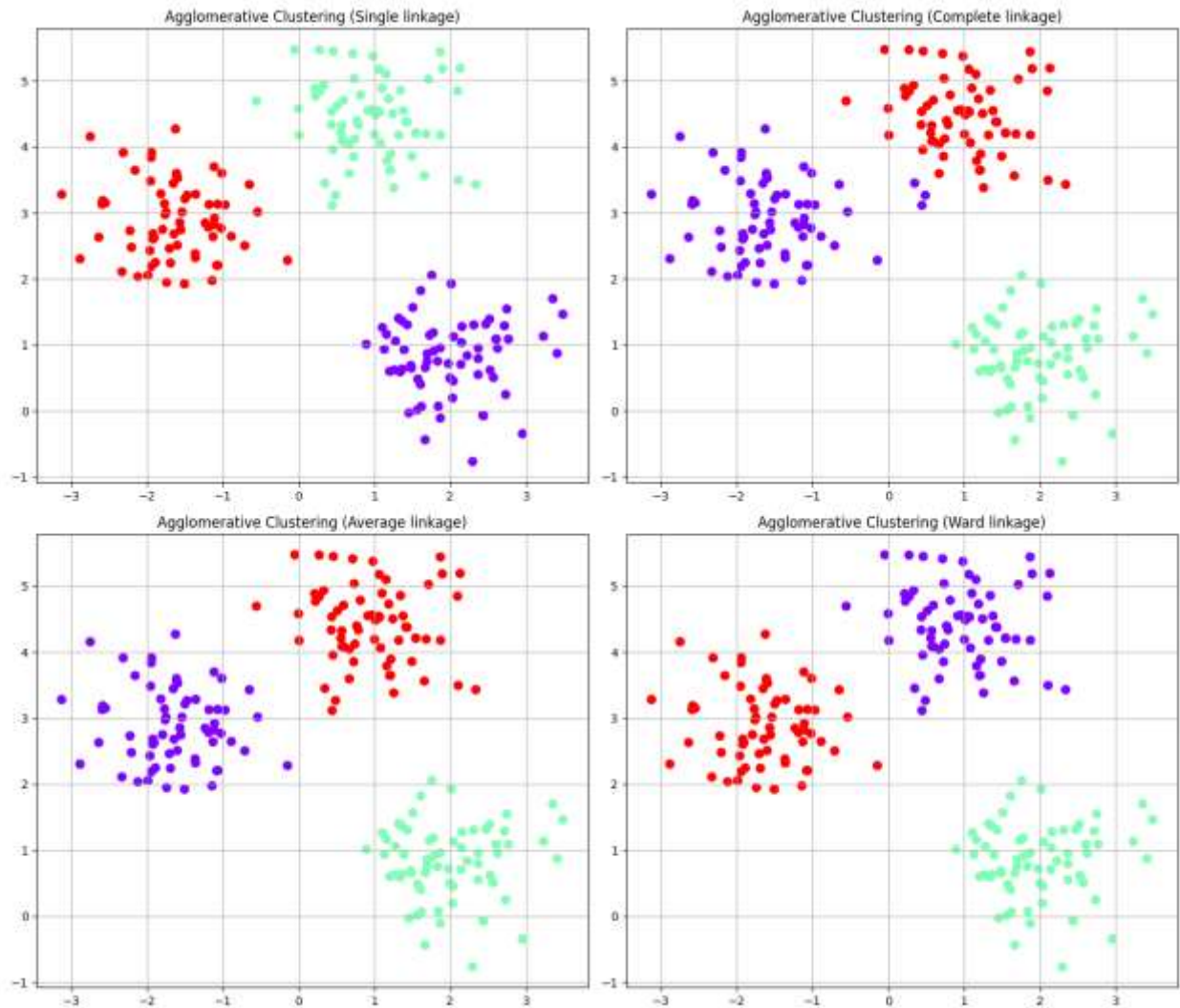


Figure 19. *Clustering results using single, complete, average, and Ward linkage methods.*

The dendrograms in Figure 20 further illustrate how each linkage method builds the cluster hierarchy. The height of the merges and the structure of the branches vary from method to method, showing how different distance interpretations influence the merging order. These differences demonstrate that the choice of linkage plays a significant role in shaping the final clustering structure, especially when the data contains subtle patterns or when cluster boundaries are not sharply defined.

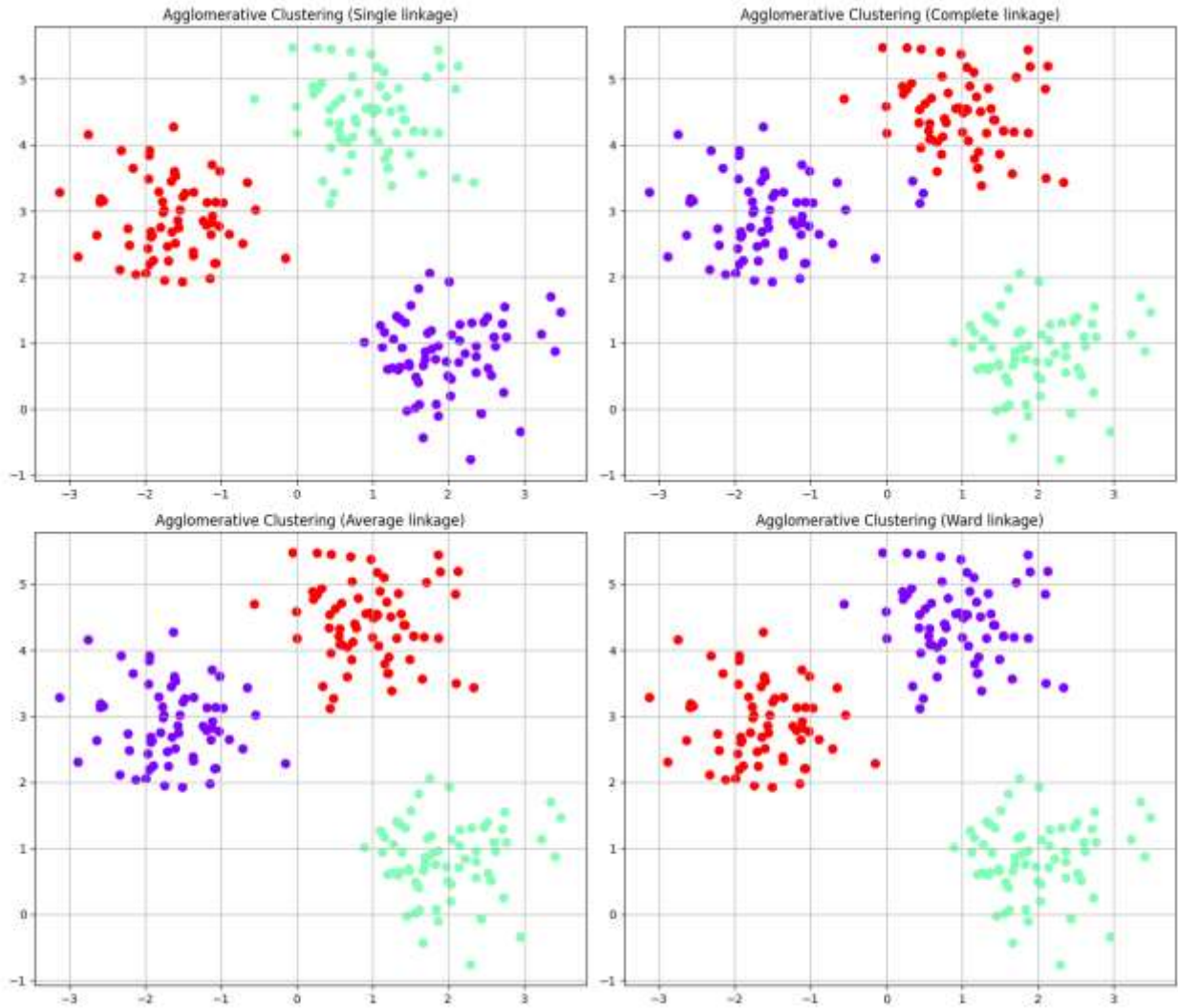


Figure 20. Dendrograms for each linkage method.

Iris Dataset (2D Projection with PCA)

The experiment used the Iris dataset to evaluate hierarchical clustering on real-world, multi-dimensional data. This dataset contains 150 flower samples with four numerical features that describe sepal and petal dimensions. Before running the clustering algorithm, the experiment scaled all features with Standard Scaler so that each variable contributed equally to the distance computations.

After scaling, the experiment applied Principal Component Analysis (PCA) and reduced the data to two principal components. This transformation preserved most of the variance in the dataset while providing a convenient two-dimensional representation for visualization. The clustering step then used agglomerative hierarchical clustering with Ward linkage and requested three clusters, which corresponds to the three known Iris species.

The figure below shows the clustering result projected onto the first two principal components. The three clusters appear as distinct groups in the PCA space, and their structure broadly matches the expected species separation. One cluster aligns clearly with *Iris setosa*, while the other two overlap more, reflecting the known similarity between *Iris versicolor* and *Iris virginica*. This behavior agrees with observations from previous labs, where algorithms also found it harder to separate these two species.

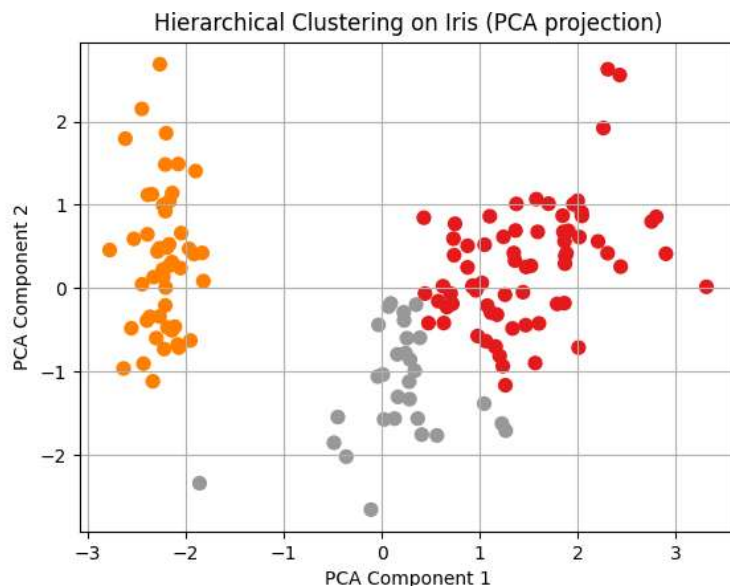


Figure 21. *Hierarchical clustering on the Iris dataset visualized in the first two principal components.*

The figure below presents the dendrogram obtained from the scaled Iris data using Ward linkage. The diagram reveals several meaningful merge levels and confirms the presence of three main groups, while also showing how individual samples join these groups at different distances. The dendrogram therefore adds an extra layer of interpretability beyond the 2D scatter plot.

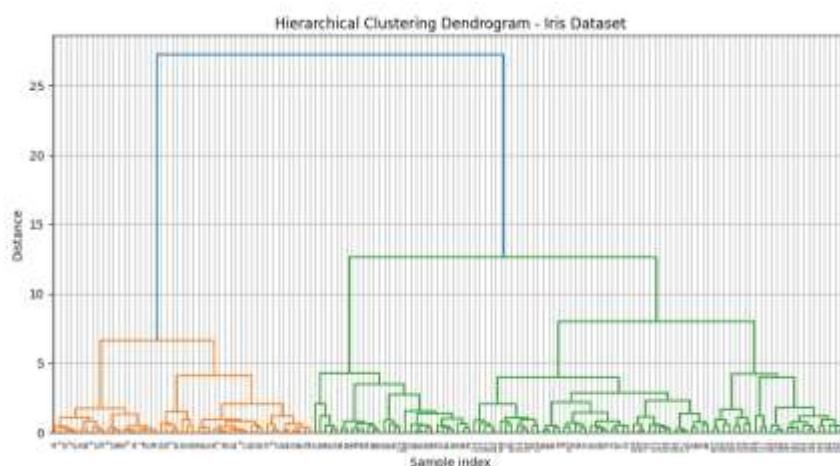


Figure 22. *Dendrogram for the Iris dataset using Ward linkage.*

Extracting Clusters Using Distance Thresholds

To explore the hierarchical structure in more detail, the experiment used the `fcluster` function to extract clusters by cutting the dendrogram at specific distances. This approach makes it possible to control the level of granularity in the clustering result. A smaller cut height produces more clusters by stopping the merging process early, while a larger cut height merges points into broader groups. The figure below shows the clusters obtained when the dendrogram was cut at a distance of 10. The algorithm returned three clusters, which aligns with the natural structure of the synthetic dataset.

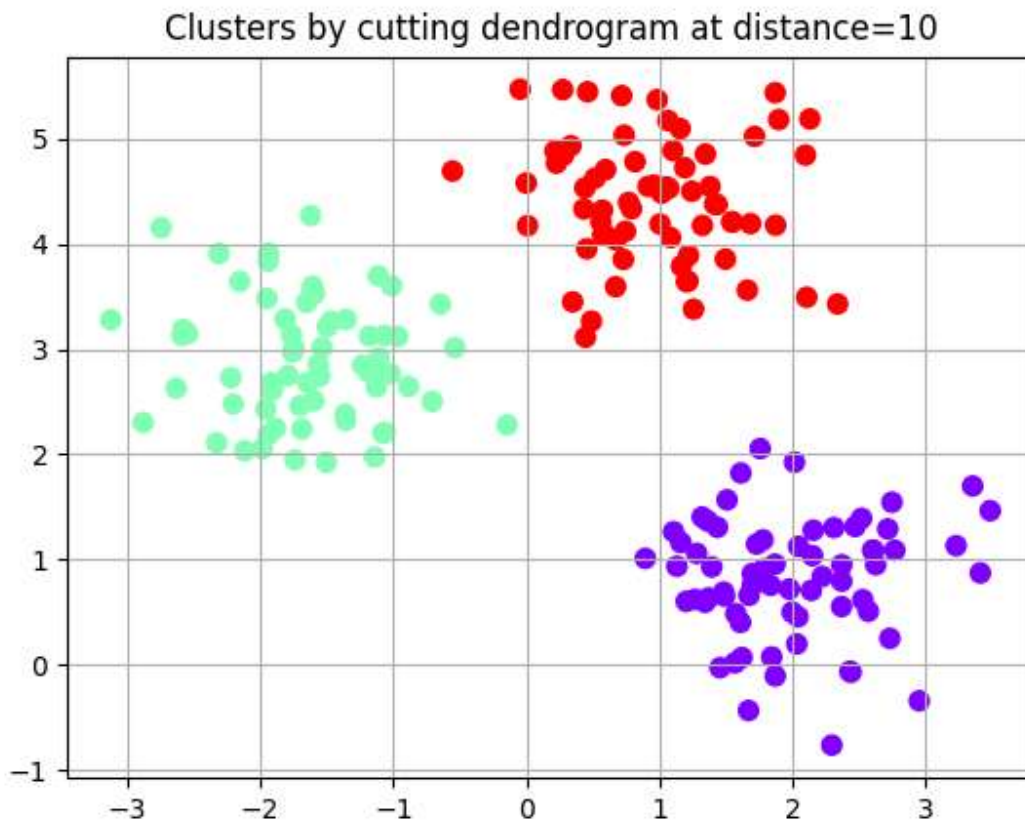


Figure 23. *Clusters extracted by cutting the dendrogram at a distance threshold of 10.*

Effect of Different Distance Thresholds on Clustering

To understand how the threshold influences the number of clusters, the experiment applied four distance values: 5, 10, 15, and 20. Each value produced a different number of clusters, ranging from several small groups at low thresholds to a nearly unified structure at high thresholds. These results show how hierarchical clustering reveals different layers of structure depending on where the dendrogram is cut. The figure below illustrates how the clustering result evolves across the four thresholds and highlights the sensitivity of the method to the choice of cut height.

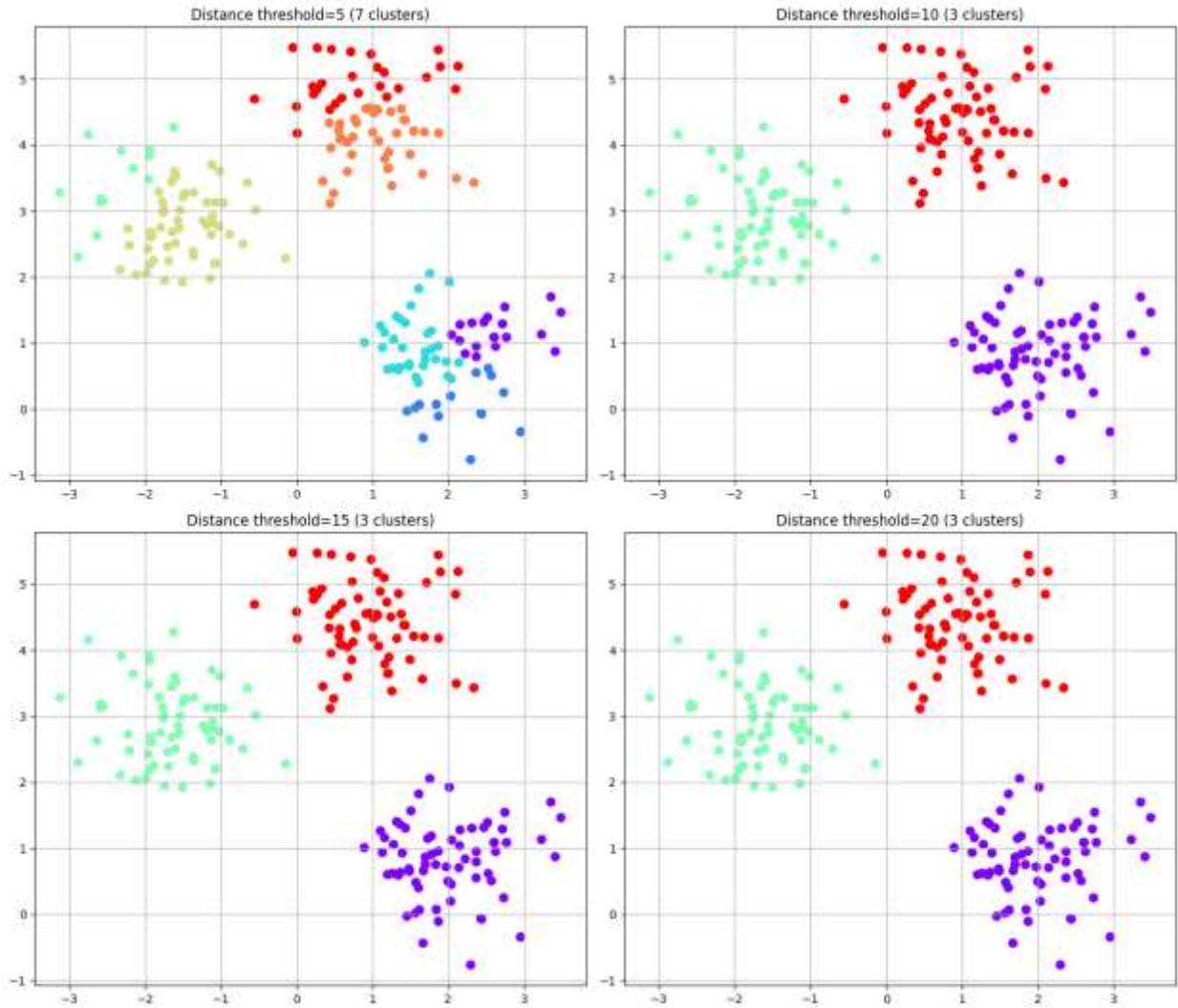


Figure 24. Clustering results for distance thresholds of 5, 10, 15, and 20.

Comparing Hierarchical Clustering with K-Means and DBSCAN

The comparison shows that each algorithm defines clusters differently and therefore produces distinct interpretations of the same dataset. K-Means works well when clusters are roughly spherical and similar in size, as it partitions the data using distances to fixed centroids. DBSCAN relies on local density, allowing it to capture arbitrary shapes and identify noise, but its performance depends strongly on the choice of parameters and can degrade when cluster densities vary. Hierarchical clustering takes a distance-based merging approach that preserves the relationships between points across multiple scales, enabling meaningful clusters to be selected directly from the dendrogram.

The side-by-side results show how each algorithm interprets the same dataset under their respective assumptions. K-Means separates the data into three compact, centroid-based clusters, which aligns well with the underlying blob structure. Hierarchical clustering with Ward linkage

produces a very similar partition, recovering the same three natural groups using a distance-based merging strategy. In contrast, DBSCAN groups all points into a single cluster and identifies no noise, indicating that the dataset does not contain strong density variations for DBSCAN to separate. This comparison illustrates how cluster geometry and density structure influence the outcome of each algorithm and why no single method behaves consistently across all datasets.

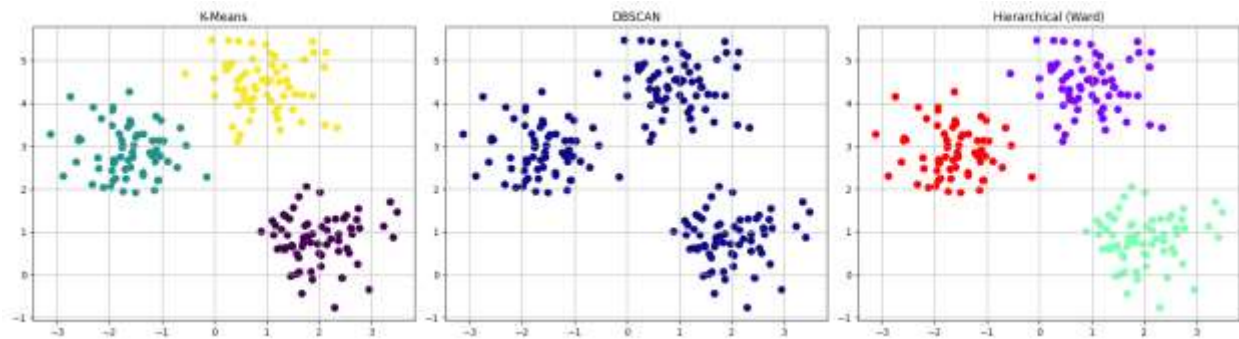


Figure 25. Comparison between K-Means, DBSCAN, and hierarchical clustering.

Questions for Discussion

1. What are the advantages of hierarchical clustering?
Hierarchical clustering works well on small to medium-sized datasets because it does not require specifying the number of clusters in advance and provides a dendrogram that reveals the full merging history, allowing you to explore the data's structure at multiple levels of granularity with a level of interpretability that many other clustering methods cannot offer.
2. When is it better to use single or Ward linkage?
Single linkage is most effective when the data contain elongated or irregular patterns that require a flexible chaining structure, whereas Ward linkage offers a stronger choice when the objective is to obtain compact, well-defined, and clearly separated clusters by controlling the growth of within-cluster variance.
3. What are the limitations compared to K-Means or DBSCAN?
Hierarchical clustering becomes increasingly expensive on large datasets because it must compute and store a full distance matrix, and unlike DBSCAN it cannot naturally detect noise or adapt to variations in density, which makes it more sensitive to outliers and less practical than methods such as K-Means or DBSCAN when the data scale or density structure becomes complex.
4. How can we automatically determine the number of clusters?
A dendrogram allows you to estimate the number of clusters by choosing a horizontal cut where large vertical gaps reveal natural separations in the data, and this decision can be supported by techniques such as inconsistency coefficients or distance thresholds, which

help identify a meaningful cut height without requiring prior knowledge of how many clusters should be extracted.

Conclusion

In conclusion, hierarchical clustering provides a powerful and interpretable framework for understanding the structure of data at multiple resolutions. Unlike K-Means, it does not require specifying the number of clusters beforehand, and unlike DBSCAN, it does not depend on density thresholds or struggle with datasets of mixed densities. Instead, it offers a complete hierarchical representation that can be explored visually through a dendrogram, making it especially valuable for exploratory data analysis and for domains where understanding relationships between clusters is as important as the clusters themselves.

However, hierarchical clustering also has limitations: it can be computationally expensive for large datasets, sensitive to noise, and influenced by the choice of linkage method. The experiments in this lab showed that Ward linkage produces compact, well-defined clusters, while other linkages such as single or complete may behave very differently depending on the data.

When compared to K-Means and DBSCAN, hierarchical clustering strikes a balance between flexibility and interpretability. K-Means remains efficient and effective for spherical, well-separated clusters; DBSCAN excels at discovering arbitrary shapes and identifying noise; and hierarchical clustering provides a multi-scale view that reveals how clusters emerge and merge. Understanding these differences is essential for selecting the most appropriate algorithm for a given dataset.

Lab 4 – Regression (Simple, Multiple, Polynomial)

Objective

The objective of this laboratory session is to study and apply regression techniques for modeling and predicting continuous numerical variables using supervised learning methods. The experiment focuses on understanding how relationships between dependent and independent variables can be quantified, interpreted, and evaluated using statistical and machine learning tools.

Specifically, this lab aims to:

- Implement Simple Linear Regression to model the relationship between a single explanatory variable and a target variable, and evaluate the strength of this relationship using error metrics and goodness-of-fit measures.
- Apply Multiple Linear Regression to analyze the combined influence of several numerical and categorical predictors on a continuous outcome and explore the use of one-hot encoding for categorical variables.
- Explore Polynomial Regression as a means of capturing nonlinear relationships, and examine how increasing model complexity affects performance and the risk of overfitting.
- Assess model quality using quantitative metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), R^2 , and Adjusted R^2 .
- Perform residual analysis to verify key regression assumptions, such as error normality and model adequacy.
- Compare linear, multiple, and polynomial models in terms of interpretability, predictive accuracy, and suitability for different types of data distributions.

Through experiments on both real-world and simulated datasets, this lab provides practical insight into the strengths, limitations, and appropriate use cases of regression models in data analysis and predictive modeling.

Theory

Regression is a supervised learning technique used to model and predict a continuous numerical variable based on one or more explanatory variables. The goal of regression analysis is to quantify the relationship between independent variables (predictors) and a dependent variable (target), and to use this relationship to make accurate predictions on unseen data.

At its core, regression seeks to determine a mathematical function that best approximates the underlying data distribution by minimizing the prediction error. In most practical settings, this is achieved using the least squares method, which minimizes the sum of squared differences between observed values and model predictions.

Simple Linear Regression

Simple linear regression models the relationship between a single independent variable x and a dependent variable y using a linear function:

$$y = b_0 + b_1x$$

where b_0 represents the intercept and b_1 represents the slope of the regression line. The parameters are estimated by minimizing the sum of squared residuals, defined as the difference between observed values and predicted values.

Simple linear regression assumes:

- A linear relationship between variables
- Independence of errors
- Homoscedasticity (constant error variance)
- Approximate normality of residuals

The quality of the model is often assessed using metrics such as Mean Squared Error (MSE) and the coefficient of determination (R^2), which measures the proportion of variance in the dependent variable explained by the model.

Multiple Linear Regression

Multiple linear regression extends the simple linear model to include multiple independent variables, allowing the combined influence of several predictors to be analyzed simultaneously:

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_px_p$$

This formulation is particularly useful when the target variable depends on several factors. However, care must be taken when adding new predictors, since the standard R^2 value never decreases as more variables are included, even if they are not relevant.

To address this limitation, the Adjusted R^2 metric is used. It introduces a penalty for unnecessary predictors and provides a more reliable measure of model quality when comparing models with different numbers of regressors.

In practical applications, datasets may include categorical variables, which cannot be directly used in linear regression. These variables must be transformed using dummy variables (one-hot encoding), allowing categorical information to be incorporated into the linear model without introducing artificial numerical orderings.

Polynomial Regression

Polynomial regression is used when the relationship between variables is nonlinear but can still be expressed as a linear combination of transformed features. Instead of fitting a straight line, the model includes polynomial terms of the independent variable:

$$y = b_0 + b_1x + b_2x^2 + \dots + b_dx^d$$

Although the model becomes nonlinear in x , it remains linear in its parameters, which allows standard linear regression techniques to be applied.

Increasing the polynomial degree generally improves the fit to the training data; however, overly complex models may capture noise rather than meaningful structure, leading to overfitting. Therefore, polynomial regression requires careful evaluation using performance metrics and residual analysis to balance model flexibility and generalization ability.

Code

The Regression Code used in this experiment is available at the following GitHub repository:

[Regression Code](#)

Simple Linear Regression (Salary Dataset)

This experiment uses the Salary dataset to illustrate the principles of simple linear regression, where a single explanatory variable is used to predict a continuous target variable. The objective is to understand how a linear regression model is built, evaluated, and validated, rather than to perform an in-depth economic analysis of salaries.

Dataset inspection and suitability

The *Salary_Data.csv* dataset is loaded using `pandas.read_csv()`. It contains 30 observations and two variables, making it well suited for demonstrating simple linear regression in a controlled and interpretable setting. The dataset consists of one explanatory variable, *YearsExperience*, and one target variable, *Salary*.

Inspection of the first rows confirms that both variables are numerical and continuous, with no missing or anomalous values. This satisfies the basic requirements of simple linear regression and ensures that parameter estimation and error analysis are not affected by data quality issues.

Descriptive statistics and linear dependency

Descriptive statistics computed using `df.describe()` summarize the central tendency and dispersion of both variables. Years of experience span from 1.1 to 10.5 years, while salaries range from €37,731 to €122,391, indicating sufficient variability to meaningfully evaluate a predictive relationship.

The correlation analysis reveals a correlation coefficient of 0.978 between the explanatory and target variables. This extremely strong positive correlation indicates that the assumption of a

linear relationship is well satisfied. In the context of simple linear regression, such a high correlation suggests that a large proportion of the variability in the target variable can potentially be explained by the predictor.

The correlation heatmap shown in **Figure 27** visually confirms this result and provides empirical justification for applying a linear regression model rather than a nonlinear alternative.

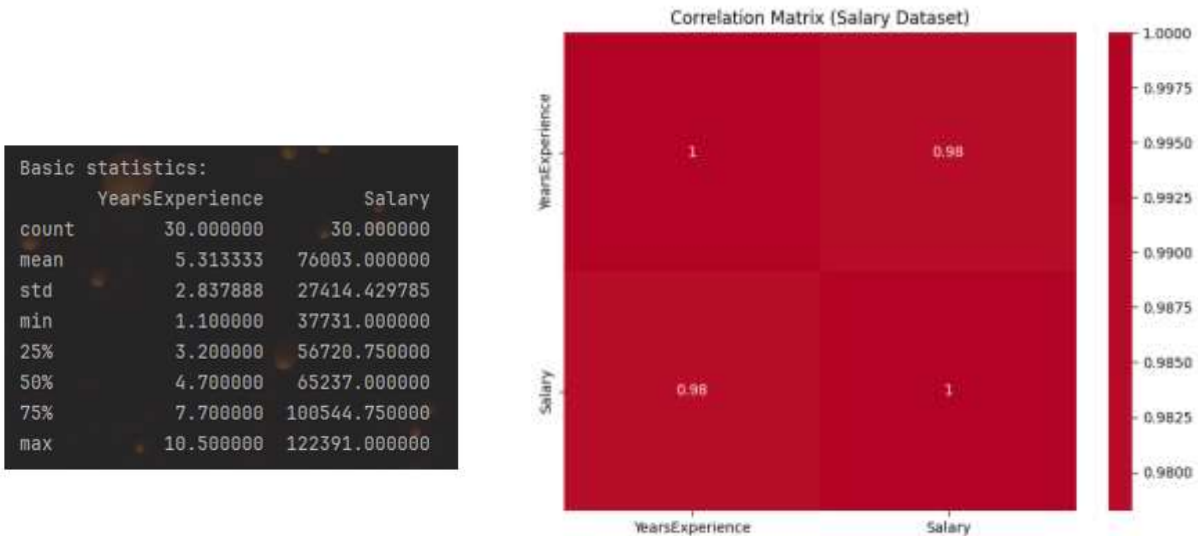


Figure 26. *Descriptive Statistics*

Figure 27. *Correlation Matrix (Salary Dataset)*

Model construction and evaluation

A simple linear regression model is constructed by assigning YearsExperience to X and Salary to y . Before training, a scatter plot is generated to visually assess the form of the relationship between the variables.

As shown in **Figure 28**, the data points follow a clear upward linear pattern with limited dispersion. This visual inspection supports the assumption of linearity and indicates that a straight-line model is appropriate for capturing the underlying relationship.



Figure 28. *Experience vs Salary*

The dataset is then split into 70% training data and 30% test data to evaluate both model fitting and generalization. The regression model is trained on the training set and evaluated on both subsets using multiple complementary metrics.

Prediction accuracy is assessed using MSE, RMSE, and MAE, which quantify the magnitude of prediction errors from different perspectives. Model explanatory power is evaluated using R^2 , which measures the proportion of variance in the target variable explained by the predictor, and Adjusted R^2 , which serves as a consistency check for model validity.

The R^2 value of 0.94 on the training set indicates that the model explains approximately 94% of the variance in the target variable. The R^2 value of 0.97 on the test set demonstrates excellent generalization, indicating that the learned linear relationship is stable and not the result of overfitting.

Adjusted R^2 values remain very close to R^2 , as expected for a model with a single explanatory variable. Error metrics further confirm strong performance: the test RMSE of approximately €4,834 and MAE of €3,737 represent relatively small errors compared to the overall scale of the target variable.

The regression results are visualized in **Figure 29** and **Figure 30**. In both cases, predicted values closely follow observed values, illustrating the ability of simple linear regression to capture the dominant linear trend in the data.

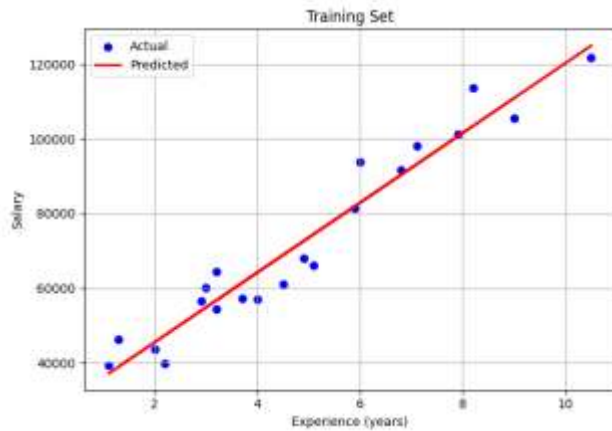


Figure 29. Training Set

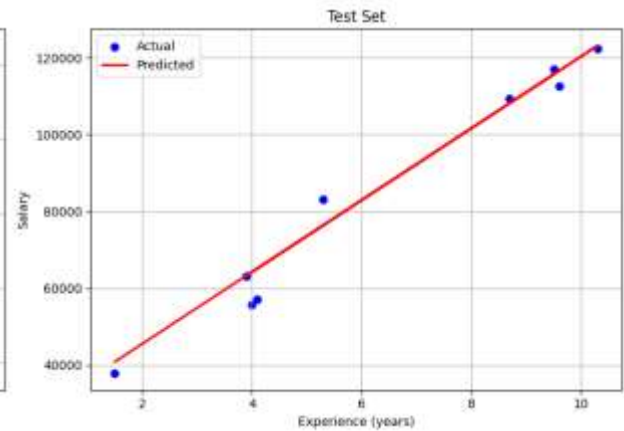


Figure 30. Test Set

Residual analysis and validation of assumptions

Residual analysis is conducted to verify the assumptions underlying simple linear regression. Residuals are computed as the difference between actual and predicted values. The mean residual for the training set is effectively zero, indicating an unbiased model. The slightly negative mean residual for the test set ($\approx -1,695$) suggests a minor underestimation tendency, which remains negligible relative to the target variable's scale.

The residual histogram in **Figure 31** shows an approximately symmetric distribution centered near zero, supporting the assumption of normally distributed errors. The Q–Q plot in **Figure 32** further confirms this observation, with most points closely aligned with the reference diagonal. Minor deviations at the extremes are typical in practical applications and do not significantly violate regression assumptions.

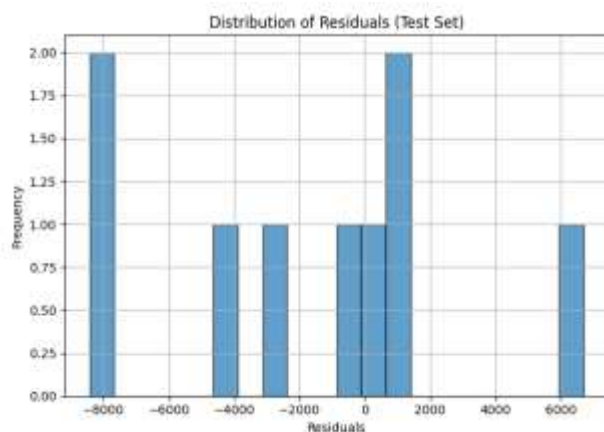


Figure 31. Residual Distribution (Test Set)

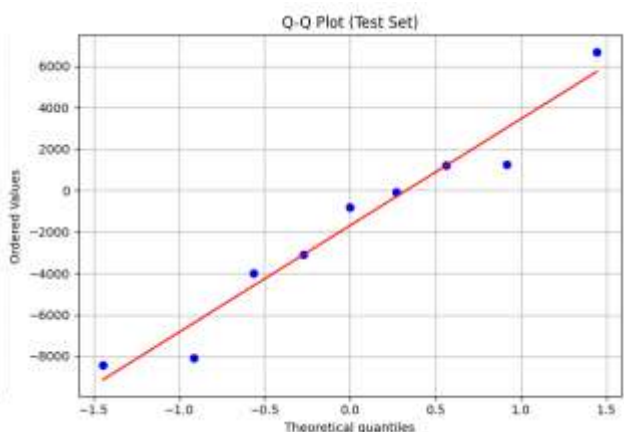


Figure 32. Q–Q Plot (Test Set Residuals)

Multiple Linear Regression (50_Startups Dataset)

This part of the experiment extends simple linear regression to multiple linear regression, where several explanatory variables are used simultaneously to predict a continuous target variable. The

objective is to understand how multiple predictors jointly contribute to a model, how categorical variables are handled, and how model evaluation changes as dimensionality increases.

Dataset inspection and structure

The dataset is loaded using `pandas.read_csv()` and contains 50 observations and five variables. The target variable is Profit, while the explanatory variables consist of three numerical predictors (R&D Spend, Administration, Marketing Spend) and one categorical predictor (State).

Inspection of the dataset confirms that all variables are complete and numerically well-defined, with no missing values. The presence of both numerical and categorical features makes this dataset particularly suitable for illustrating real-world preprocessing requirements in multiple linear regression.

Descriptive statistics and correlation analysis

Descriptive statistics indicate that the numerical predictors operate on markedly different scales. R&D Spend and Marketing Spend show large ranges and high variability, whereas Administration varies within a narrower interval. This difference in scale and dispersion is common in real-world datasets and highlights that predictors may contribute unequally to a regression model.

The correlation matrix shown in **Figure 33** clarifies the linear relationships between each predictor and the target variable. R&D Spend exhibits a very strong positive correlation with Profit (0.97), indicating that it is the primary driver of the model's explanatory power. Marketing Spend shows a moderate positive correlation (0.75), suggesting a secondary contribution, while Administration displays a weak correlation (0.20), implying a limited direct linear effect on profit.

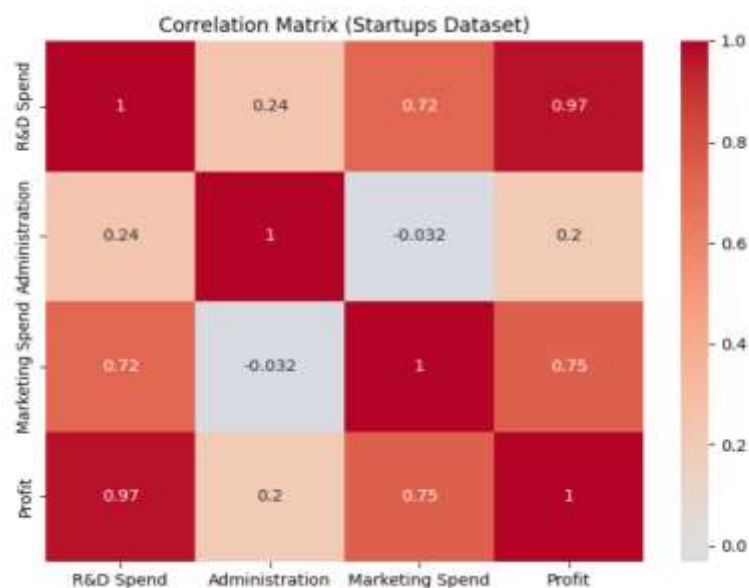


Figure 33. *Correlation Matrix (50_Startups Dataset)*

The matrix also reveals a substantial correlation between R&D Spend and Marketing Spend (0.72). This interdependence illustrates an important characteristic of multiple linear regression: predictors are not necessarily independent. Such correlations between explanatory variables can influence coefficient estimates and complicate interpretation, making global performance metrics and residual analysis essential for reliable model evaluation.

Exploratory visualization of predictors

To further examine individual predictor–target relationships, scatter plots are generated for each numerical feature against Profit. As shown in **Figures 34–36**, R&D Spend displays a strong linear trend, Marketing Spend shows a weaker but still visible relationship, and Administration exhibits substantial scatter with no clear linear structure.

These visualizations reinforce the correlation analysis and demonstrate why multiple regression is necessary: while one variable may dominate, additional predictors can still contribute explanatory power when combined appropriately.

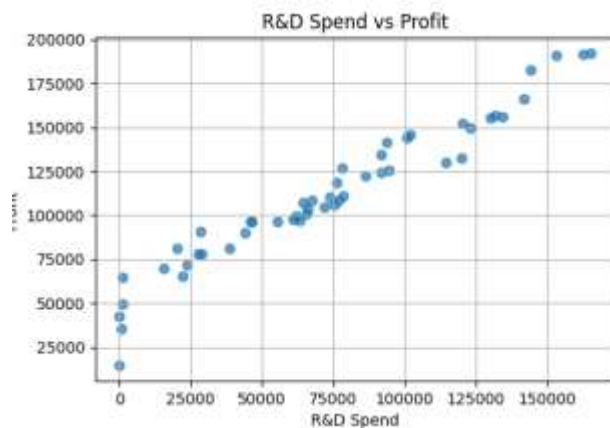


Figure 34. *R&D Spend vs Profit*

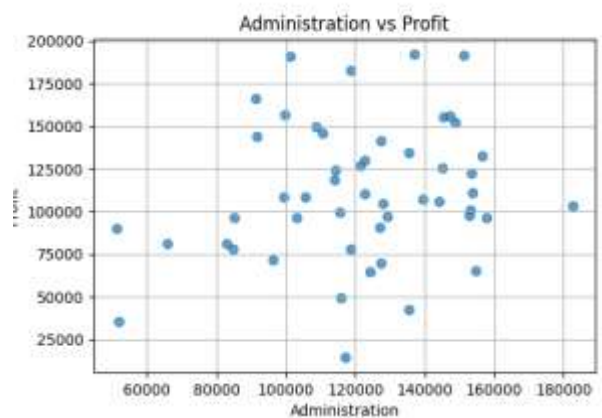


Figure 35. *Administration vs Profit*

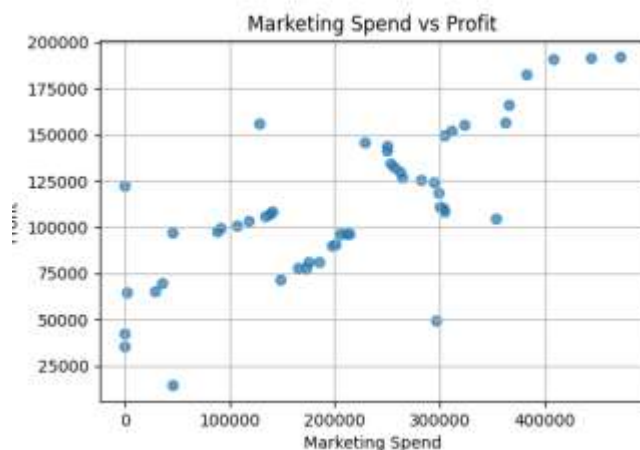


Figure 36. *Marketing Spend vs Profit*

Handling categorical variables

Multiple linear regression requires numerical input features. Since State is a categorical variable, it cannot be directly used in the regression model. To address this, one-hot encoding is applied using a ColumnTransformer combined with OneHotEncoder.

This transformation converts the categorical variable into a set of binary dummy variables, allowing the model to account for state-specific effects without imposing any artificial numerical ordering. All numerical predictors are passed through unchanged, resulting in a fully numerical feature matrix suitable for regression.

Model training and performance evaluation

After preprocessing, the dataset is split into 70% training data and 30% test data. A multiple linear regression model is trained on the training set and evaluated on both subsets using error-based and goodness-of-fit metrics.

The training R^2 value of 0.95 indicates that approximately 95% of the variance in profit is explained by the combined predictors. The test R^2 value of 0.94 confirms strong generalization to unseen data, indicating that the model captures stable linear relationships rather than noise.

In contrast to simple linear regression, Adjusted R^2 plays a critical role here. The Adjusted R^2 decreases from 0.94 (training) to 0.89 (test), reflecting the penalty applied for increased model complexity. This reduction indicates that while the additional predictors improve explanatory power, not all contribute equally, and some may add limited information beyond the strongest variables.

Error metrics further support these conclusions. The test MSE of approximately 61.9 million reflects larger absolute errors than in the simple regression case, which is expected due to the broader scale and variability of the target variable. Nevertheless, the consistency between training and test errors indicates that the model is not overfitting. The comparison between actual and predicted profits on the test set is shown in **Figure 37**, where predicted values closely track observed values across sample

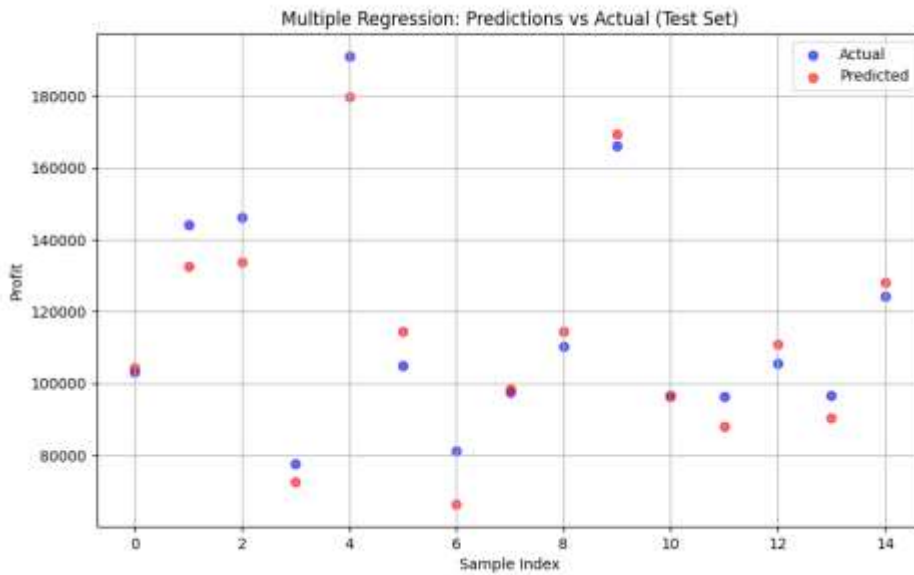


Figure 37. *Multiple Linear Regression — Actual vs Predicted (Test Set)*

Residual analysis and model validation

Residuals are computed on the test set to verify regression assumptions. The residual histogram shown in **Figure 38** is approximately symmetric and centered around zero, suggesting that the model does not exhibit systematic bias.

The Q–Q plot in **Figure 39** shows that residuals largely follow a normal distribution, with minor deviations at the extremes. Such deviations are common in real-world datasets and do not significantly undermine the validity of the model.

Together, these diagnostics indicate that the assumptions of multiple linear regression are reasonably satisfied.

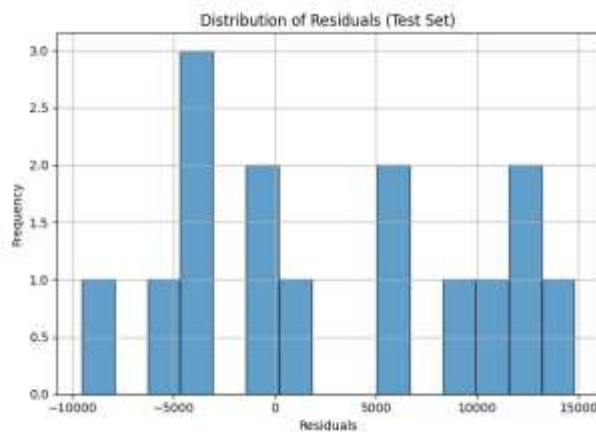


Figure 38. *Residual Distribution (Test Set)*

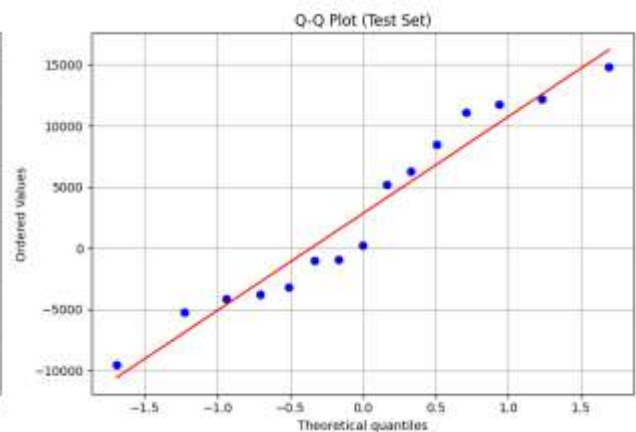


Figure 39. *Q–Q Plot (Test Set Residuals)*

Example prediction and practical interpretation

To illustrate the practical use of the multiple linear regression model, an example prediction is performed using a hypothetical startup configuration. After applying one-hot encoding to the categorical variable *State*, a new input representing a startup located in New York with an R&D Spend of €130,000, Administration costs of €140,000, and Marketing Spend of €300,000 is provided to the trained model.

The model predicts a profit of approximately €158,745 for this configuration. This result demonstrates how multiple linear regression combines the weighted contributions of several predictors to generate a single numerical estimate. Unlike simple regression, the predicted value reflects the simultaneous influence of multiple operational factors rather than reliance on a single explanatory variable.

```
# Example prediction (after one-hot encoding)
sample_input = [[1, 0, 0, 130000, 140000, 300000]] # [NewYork, California, Florida, R&D, Admin, Marketing]
prediction = regressor_multi.predict(sample_input)
print(f"\nPrediction for input {sample_input}: ${prediction[0]:,.2f}")

Prediction for input [[1, 0, 0, 130000, 140000, 300000]]: $158,745.12
```

Figure 40. *Snapshot of python code and response*

Polynomial Regression

Polynomial regression is applied in this experiment to model relationships where changes in the target variable accelerate or decelerate as the input increases. Unlike linear models, polynomial regression allows the rate of change itself to vary, making it suitable for datasets where growth is clearly non-uniform across the input range.

This section examines polynomial regression in two contexts:

- A structured dataset illustrating hierarchical salary progression.
- A simulated nonlinear dataset used to analyze model behavior, complexity, and alternative modeling strategies.

Polynomial Regression on the Position Salaries Dataset

Data structure and nonlinear behavior

The **Position_Salaries.csv** dataset describes salary as a function of hierarchical position level. When visualized (Figure 41), the data exhibits a clear nonlinear pattern: salary increases slowly at lower levels and rises sharply at senior levels. This structure reflects a common real-world phenomenon where promotions at higher ranks produce disproportionately larger compensation increases. This curvature indicates that a constant-rate model is inadequate and motivates the use of polynomial regression to capture changes in growth intensity across position levels.

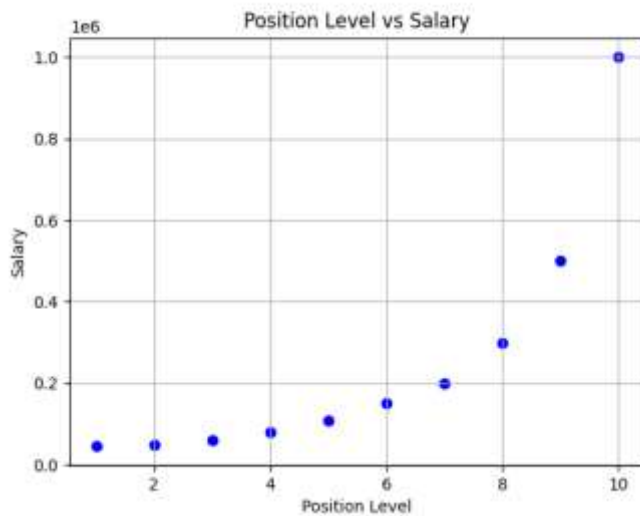


Figure 41. *Position Level vs Salary*

A linear regression model is first fitted to serve as a baseline. As shown in **Figure 42**, the fitted line systematically underestimates salaries at higher position levels and overestimates them at lower levels. This mismatch is not caused by noise, but by the model's inability to represent accelerating growth.

The relatively low explanatory performance confirms that the relationship is not well approximated by a constant slope, reinforcing the need for a more flexible functional form.

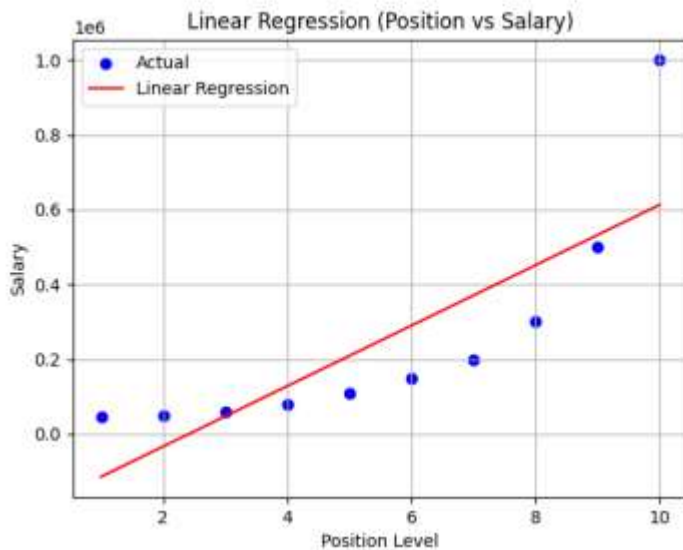


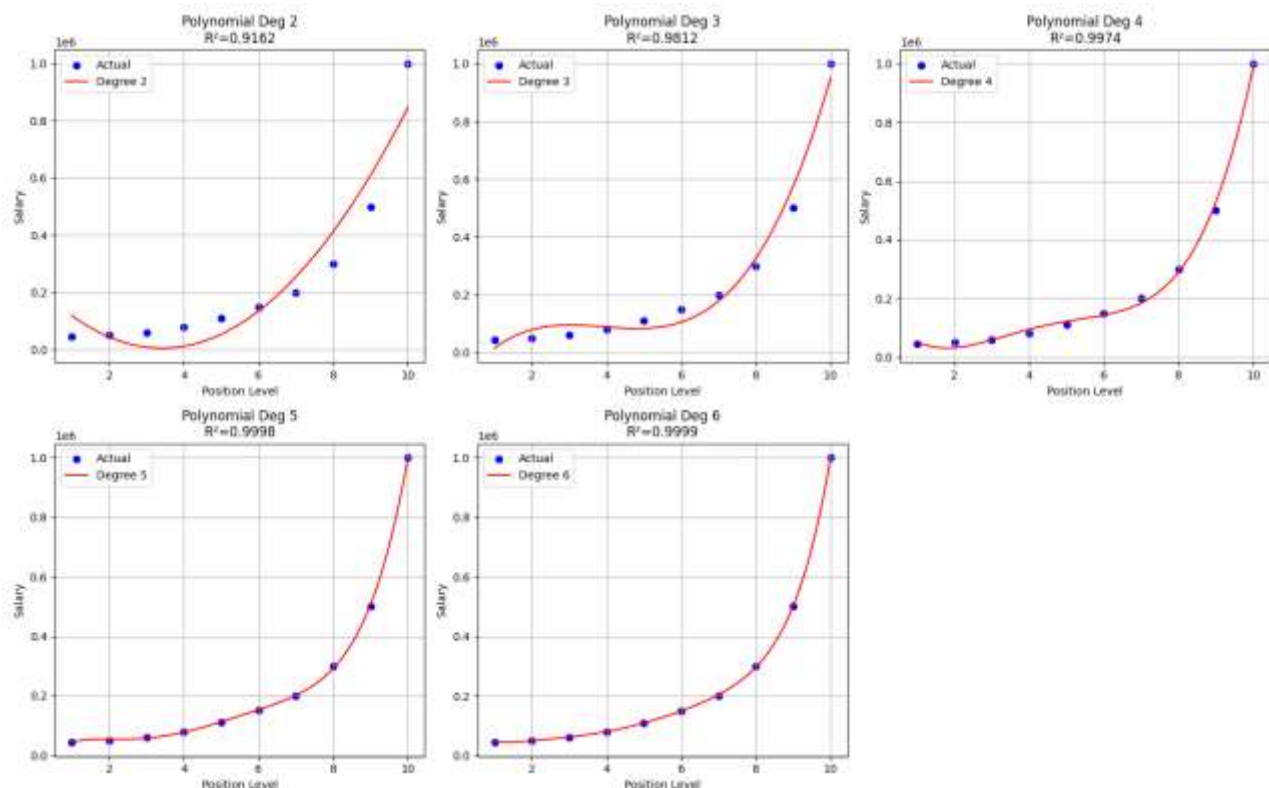
Figure 42. *Linear Regression (Position vs Salary)*

Polynomial regression models of degrees 2 through 6 are then fitted to the data. As illustrated in **Figures 43–47**, increasing the polynomial degree progressively improves the model’s ability to follow the observed salary trajectory.

The following observations were made:

- Lower-degree models capture basic curvature but still underestimate rapid salary growth.
- Intermediate degrees provide a strong balance between smoothness and accuracy.
- Higher-degree models closely interpolate the data points, achieving near-perfect fit.

This progression demonstrates how polynomial regression adapts model flexibility by increasing degree, allowing the fitted curve to reflect nonlinear trends that are structurally present in the data.



Figures 43–47. Polynomial Regression Models (Degrees 2–6)

The improvement in R^2 values with increasing polynomial degree confirms that higher-order models explain a larger proportion of variance. However, this improvement must be interpreted cautiously. The dataset contains a small number of observations, and higher-degree models risk learning the specific data configuration rather than a generalizable pattern.

Adjusted R^2 values provide insight into this trade-off. While they remain high, gains diminish as degree increases, indicating that additional complexity yields progressively smaller explanatory benefits.

Polynomial Regression on Simulated Nonlinear Data

To further analyze polynomial behavior, a synthetic dataset is generated where purchase amount decreases proportionally to the square of page loading time. The resulting scatter plot (Figure 48) exhibits a steep nonlinear decay, with large values at short loading times and rapid convergence toward zero.

This dataset is intentionally constructed to challenge polynomial regression, as the underlying relationship is not naturally polynomial.

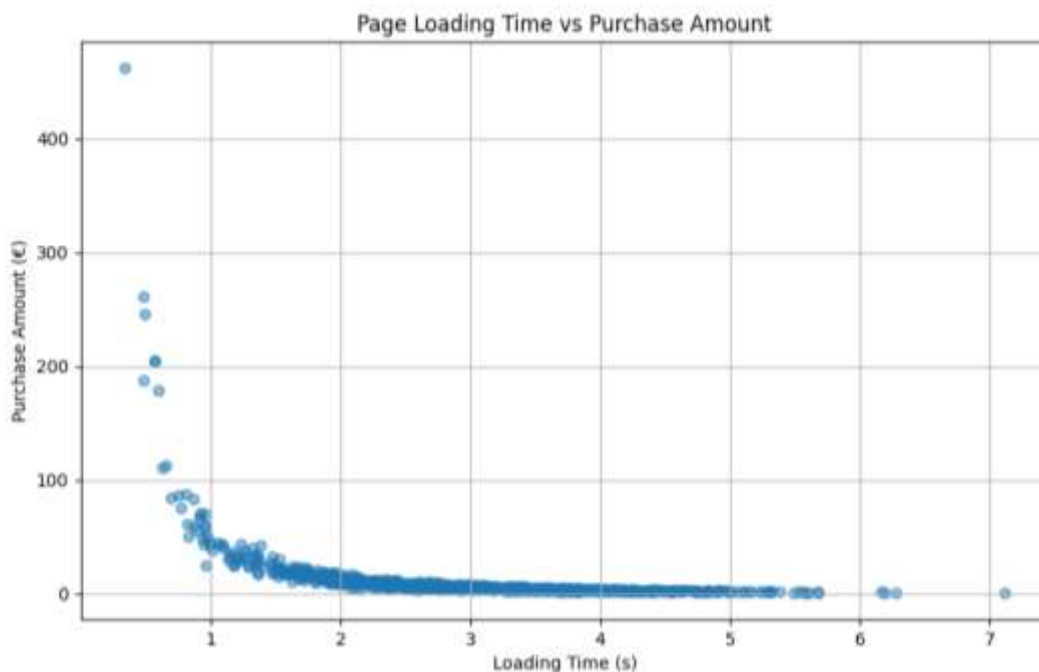
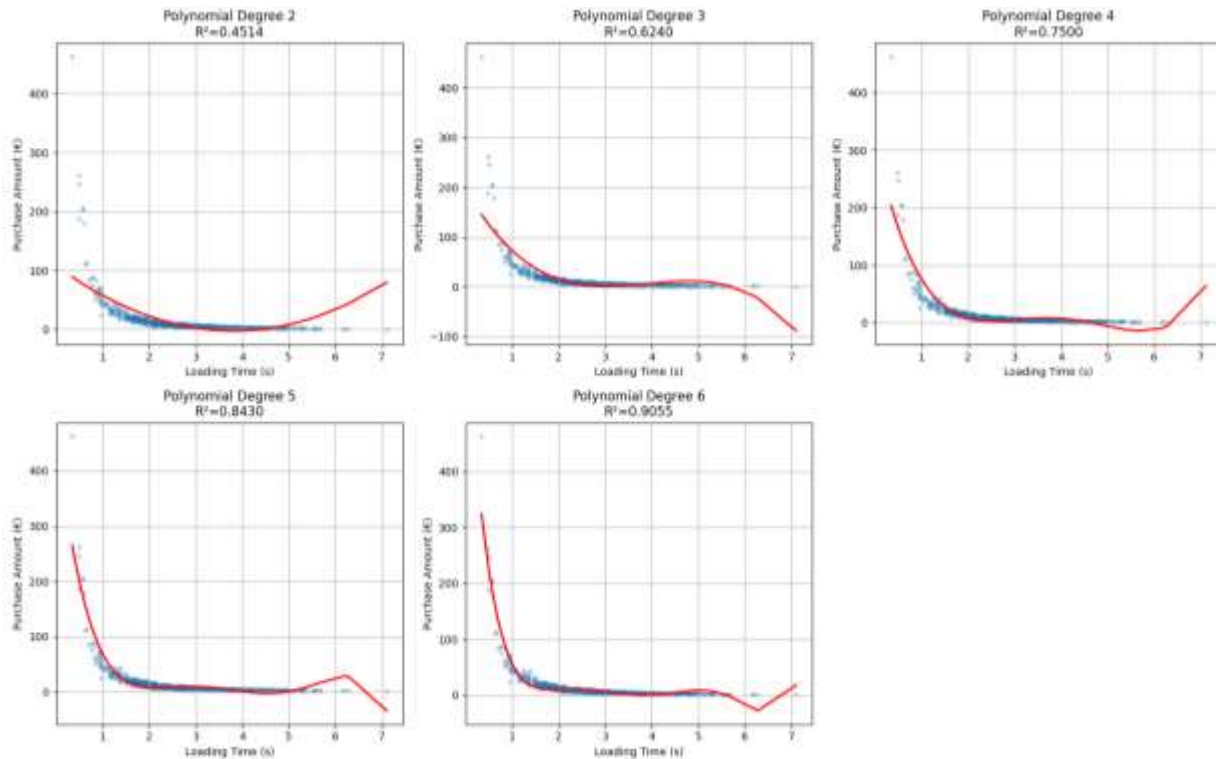


Figure 48. *Page Loading Time vs Purchase Amount*

Polynomial approximation and instability

Polynomial models of increasing degree are fitted to the simulated data, as shown in **Figures 49–53**. Although higher-degree models achieve improved R^2 values, the fitted curves display oscillatory behavior and instability, particularly near the boundaries of the input range.

These artifacts illustrate a key limitation of polynomial regression: when the functional form does not match the true relationship, increasing degree can improve numerical fit while degrading interpretability and stability.



Figures 49–53. *Polynomial Regression on Simulated Data (Degrees 2–6)*

Feature transformation and model reinterpretation

To address this issue, the explanatory variable is transformed using a $1/x^2$ mapping, reflecting the known structure of the data-generating process. After transformation, a simple linear regression model is applied. As shown in **Figure 54**, the transformed model achieves a high R^2 value (≈ 0.97) while producing a smooth and stable fit without oscillations. This result demonstrates that appropriate feature transformation can outperform high-degree polynomial models, both in accuracy and robustness.

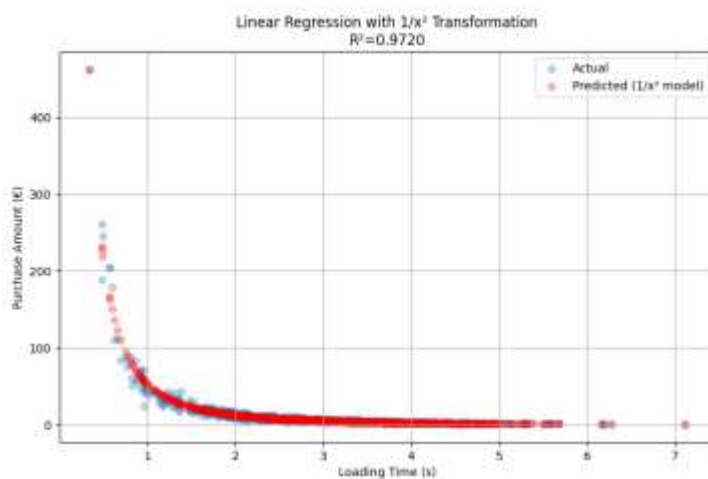


Figure 54. *Linear Regression with $1/x^2$ Transformation*

Conclusion

This laboratory successfully demonstrated how regression models can be constructed, evaluated, and validated for predicting continuous numerical outputs using supervised learning. Across all parts of the experiment, the main focus was not the datasets themselves, but how regression methods behave under different data structures and modeling requirements.

In the simple linear regression experiment, the lab showed how a single predictor can be used to learn an interpretable linear relationship and produce accurate predictions when the linearity assumption is satisfied. The strong goodness-of-fit values and small error metrics confirmed that the model captured the dominant trend effectively, while residual diagnostics supported the validity of key assumptions such as unbiased errors and approximate normality.

The multiple linear regression experiment expanded this approach to a realistic setting with several predictors and a categorical feature. This part further emphasized practical preprocessing using one-hot encoding and demonstrated how multiple variables jointly contribute to prediction. The strong test performance confirmed that the model generalized well, while the drop in Adjusted R^2 highlighted an important methodological point: adding predictors increases complexity, and not all variables necessarily provide meaningful additional explanatory power. Residual analysis further supported the suitability of the linear model for the startups dataset.

Finally, polynomial regression illustrated how nonlinear patterns can be captured by introducing higher-degree feature transformations. In the Position Salaries dataset, increasing polynomial degree improved model fit by matching the accelerating salary trend. However, the experiment also made clear that better fit does not automatically mean better modeling, especially with small datasets where high-degree polynomials can become overly sensitive and risk overfitting. This limitation became even clearer in the simulated nonlinear dataset: polynomial models improved with degree but produced unstable curves, particularly at the boundaries. The transformation-based approach ($1/x^2$) achieved superior accuracy with a simpler and more stable model, reinforcing a key lesson: selecting a representation that matches the underlying relationship often outperforms increasing model complexity.

Lab 5 – Logistic Regression

Objective

The objective of this laboratory session is to study and apply logistic regression as a supervised learning technique for binary classification problems. The experiment focuses on understanding how probabilistic models can be used to separate data into two classes, evaluate classification performance, and interpret decision boundaries in feature space.

Specifically, this lab aims to:

- Implement logistic regression to model the probability of a binary outcome based on numerical input features.
- Understand the role of feature scaling in gradient-based classification models.
- Analyze how logistic regression estimates class probabilities and converts them into discrete class predictions using decision thresholds.
- Evaluate classification performance using accuracy, precision, recall, F1-score, and confusion matrices.
- Visualize the learned decision boundary in a two-dimensional feature space for both training and test datasets.
- Assess model quality using Receiver Operating Characteristic (ROC) curves and the Area Under the Curve (AUC) metric.
- Interpret the trade-off between false positives and false negatives through probabilistic predictions.

Through experiments on a real-world social network advertising dataset, this lab provides practical insight into how logistic regression operates as a baseline classification model, its strengths in interpretability and efficiency, and its limitations when class boundaries are not linearly separable.

Theory

Logistic regression is a supervised learning method for binary classification, where the target variable takes one of two possible values, commonly encoded as 0 and 1. Instead of predicting a continuous numerical outcome, logistic regression models the probability that an observation belongs to a particular class, making it especially suitable for decision-oriented tasks.

At its core, logistic regression constructs a linear combination of input features. Given a feature vector

$$X = (x_1, x_2, \dots, x_p),$$

the model computes a linear score:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

This linear score is then transformed using the sigmoid (logistic) function, which maps any real-valued input to the interval $[0, 1]$:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The output of the sigmoid function is interpreted as a probability estimate:

$$P(y = 1 | X) = \sigma(z)$$

This probabilistic formulation allows logistic regression to express classification confidence, rather than producing only hard class assignments.

Decision rule

To convert predicted probabilities into class labels, a decision threshold is applied. By convention, a threshold of 0.5 is used:

- If $P(y = 1 | X) \geq 0.5$, the model predicts class 1
- Otherwise, it predicts class 0

The threshold can be adjusted depending on the application, enabling control over the trade-off between **false positives** and **false negatives**. This flexibility is particularly important in domains where the costs of misclassification are asymmetric.

Model training and optimization

The parameters of logistic regression are estimated by **maximizing the likelihood** of the observed data. In practice, this is equivalent to **minimizing the log-loss (cross-entropy loss)**, which strongly penalizes confident but incorrect predictions.

Unlike linear regression, logistic regression does not admit a closed-form solution for its parameters. Instead, optimization is performed using **iterative numerical methods**, such as gradient descent or more advanced solvers implemented in machine learning libraries.

Feature scaling and interpretability

Because logistic regression relies on gradient-based optimization, feature scaling is important for stable and efficient convergence. Standardization ensures that all input features contribute proportionally to parameter updates and prevents dominance by variables with larger numerical ranges.

The learned coefficients remain highly interpretable. Each coefficient represents the effect of the corresponding feature on the log-odds of the positive class, assuming all other features remain constant. This interpretability makes logistic regression a preferred model when transparency and explainability are required.

Model evaluation for classification

Evaluation of logistic regression differs from that of regression models, as performance must be assessed in terms of **classification quality** rather than numerical error. Common evaluation tools include:

- **Accuracy**, measuring overall correctness
- **Precision and recall**, capturing class-specific performance
- **Confusion matrices**, summarizing prediction outcomes
- **ROC curves**, evaluating performance across all decision thresholds
- **Area Under the Curve (AUC)**, measuring the model's ability to discriminate between classes independently of a fixed threshold

Together, these metrics provide complementary perspectives on model behavior and decision reliability.

Code

The Logistic Regression Code used in this experiment is available at the following GitHub repository: [Logistic Regression Code](#)

Implementation using Social Network Ads Dataset

This experiment applies logistic regression to a binary classification problem: predicting whether a user purchases a product ($\text{Purchased} = 1$) or not ($\text{Purchased} = 0$) based on two numerical features, Age and Estimated Salary. The goal is to demonstrate the full logistic-regression pipeline: selecting appropriate inputs, scaling features, training a probabilistic classifier, and interpreting results using classification-specific evaluation tools.

Dataset inspection and modeling setup

The *Social_Network_Ads.csv* dataset is first examined to verify its suitability for binary classification. It contains 400 observations and five variables, including demographic information and a binary outcome variable indicating whether a purchase was made.

For this experiment, Age and Estimated Salary are selected as explanatory variables, while Purchased is used as the target variable. The target is already encoded as a binary label (0 = not purchased, 1 = purchased), which aligns directly with the probabilistic formulation of logistic regression. Restricting the model to two numerical predictors enables direct visualization of the learned decision boundary and facilitates geometric interpretation of the classifier in feature space.

Feature scaling

Before training, both predictors are standardized using StandardScaler. This step is essential because EstimatedSalary has a much larger numeric scale than Age. Without scaling, the optimization procedure would be dominated by the salary feature's magnitude, slowing

convergence and potentially producing a poorly balanced boundary. Scaling ensures that the learned weights reflect *real predictive contribution* rather than the units of measurement.

Model training and probabilistic predictions

The logistic regression model is trained on the standardized training set. Instead of predicting a continuous value, the model learns a linear decision function in feature space and converts it into a probability of purchase. Final class predictions are obtained by applying a default 0.5 threshold (probability $\geq 0.5 \rightarrow$ Purchased = 1). This probability-based mechanism is important because it allows the decision rule to be adjusted later depending on whether the application prefers fewer false alarms or fewer missed buyers.

Evaluation of results

The logistic regression model achieves an overall accuracy of 81% on the test set, indicating that a substantial majority of users are classified correctly. However, accuracy alone does not fully describe classifier behavior, particularly when the costs of different error types are asymmetric. For this reason, a more detailed evaluation is conducted using class-specific metrics and the confusion matrix.

Performance differs noticeably between the two classes. For non-buyers (class 0), the model exhibits high precision (0.82) and very high recall (0.91), demonstrating strong capability in identifying users who do not make a purchase. In practice, this means that most non-buyers are correctly rejected, and predictions of “not purchased” are reliable. In contrast, for buyers (class 1), the model achieves good precision (0.79) but moderate recall (0.64). This indicates that while predictions of a purchase are usually correct, a significant proportion of actual buyers are not detected by the model.

This imbalance reveals a conservative classification strategy: the model tends to favor predicting the negative class when uncertainty exists, thereby reducing false positives at the expense of increasing false negatives. Such behavior is common in purchase prediction tasks, where buyers and non-buyers often overlap in feature space and decision boundaries cannot perfectly separate the two groups.

```
=== Accuracy ===
0.8100

=== Classification report ===
```

	precision	recall	f1-score	support
0	0.82	0.91	0.86	64
1	0.79	0.64	0.71	36
accuracy			0.81	100
macro avg	0.81	0.77	0.78	100
weighted avg	0.81	0.81	0.80	100

Figure 55. *Evaluation of Logistic Regression*

The confusion matrix in **Figure 56** provides a concrete breakdown of these outcomes. The model correctly classifies 58 non-buyers and 23 buyers, while producing 6 false positives and 13 false negatives. The higher number of false negatives confirms the lower recall for the positive class and highlights the model's tendency to miss some true purchasers rather than incorrectly label non-buyers as buyers.

From an application perspective, this error structure has meaningful implications. Missing potential buyers corresponds to lost targeting opportunities, while false positives correspond to unnecessary marketing effort. The current model prioritizes minimizing wasted outreach over maximizing buyer detection. Whether this trade-off is desirable depends on the relative costs of these errors in the specific business context. Importantly, logistic regression's probabilistic output allows this balance to be adjusted by modifying the decision threshold, a point further explored through ROC analysis.

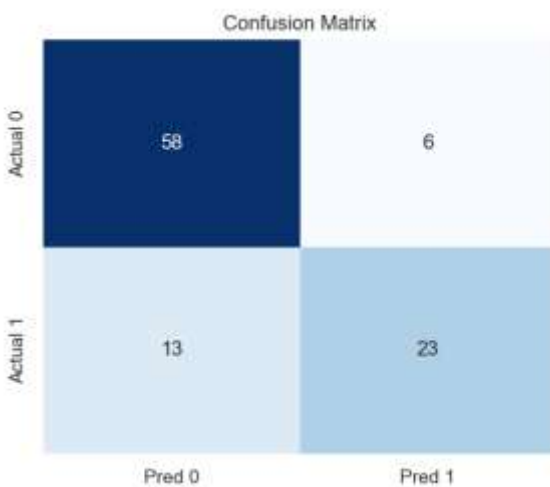


Figure 56. *Confusion Matrix*

Decision boundary visualization

The decision boundary visualizations in **Figures 57** and **58** illustrate how the logistic regression model separates the feature space defined by Age and Estimated Salary into two decision regions corresponding to the predicted classes.

In both the training and test sets, the boundary divides the space into a low-probability region classified as *Not Purchased* and a high-probability region classified as *Purchased*. Observations close to the boundary represent cases where small changes in age or salary lead to different predicted outcomes. The presence of both classes in this narrow transition zone reflects genuine overlap in user behavior.

Importantly, the decision boundary maintains a similar orientation and position across the training and test plots. This consistency indicates that the model has learned a stable linear separation rather than fitting noise specific to the training data.

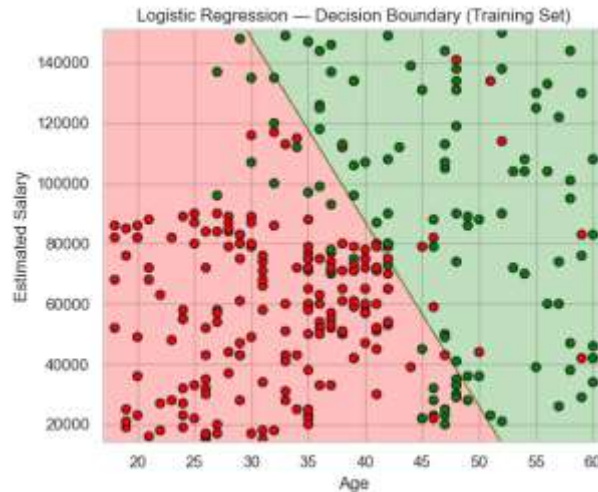


Figure 57. *Decision Boundary (Training Set)*

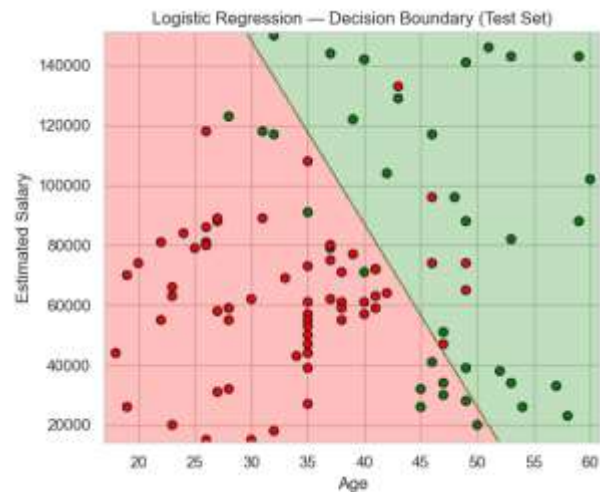


Figure 58. *Decision Boundary (Test Set)*

ROC curve and AUC (threshold-independent evaluation)

The Receiver Operating Characteristic (ROC) curve provides a threshold-independent evaluation of classifier performance by examining the trade-off between the true positive rate and the false positive rate across all possible probability thresholds. Unlike accuracy or confusion matrices, which depend on a fixed decision threshold, the ROC curve assesses how well the model ranks positive instances above negative ones.

The Area Under the Curve (AUC) value of **0.913** indicates strong discriminative power. This means that, in the majority of cases, the model assigns higher predicted probabilities to users who actually purchase than to those who do not. Such a value reflects a well-separated probability distribution between the two classes, even when their feature values partially overlap.

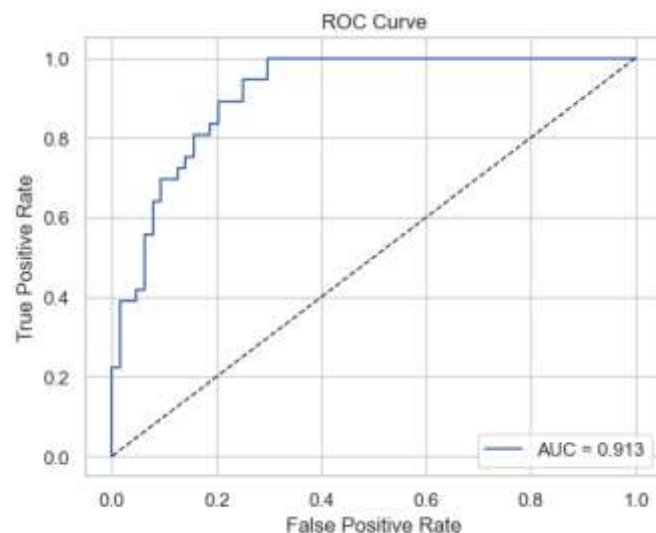


Figure 58. *ROC Curve (AUC = 0.913)*

The contrast between the AUC computed using predicted probabilities (0.913) and the AUC derived from hard class labels (0.773) is particularly informative. Hard labels discard probability information by enforcing a single threshold, whereas ROC/AUC preserves the full ranking structure of predictions.

```
AUC (using predicted probabilities): 0.9128
AUC (using hard labels only):      0.7726
```

Figure 59. *Contrast between the AUC computed using probabilities and hard-class labels*

Conclusion

This laboratory session demonstrated the practical application of logistic regression as a probabilistic model for binary classification. Using the Social Network Ads dataset, logistic regression was employed to predict purchase behavior based on age and estimated salary, providing a clear and interpretable framework for understanding classification decisions in a two-dimensional feature space.

The experiment showed that logistic regression effectively learns a linear decision boundary that separates buyers from non-buyers while producing meaningful probability estimates rather than only hard class labels. The use of feature standardization was essential for stable optimization, ensuring that both predictors contributed appropriately during model training.

Model evaluation highlighted the importance of analyzing classification performance beyond overall accuracy. Although the classifier achieved an accuracy of 81%, deeper inspection through the confusion matrix and classification report revealed an asymmetric error pattern. The model performed very well at identifying non-buyers, while exhibiting lower recall for buyers, indicating a conservative decision behavior that favors reducing false positives at the cost of missing some true purchasers. This trade-off reflects the inherent overlap between classes in the Age–Salary space rather than a modeling error.

Decision boundary visualizations confirmed that the learned separation is stable across training and test datasets, suggesting good generalization and no evidence of overfitting. The ROC curve and the high AUC value (≈ 0.91) further demonstrated that the model possesses strong discriminative capability across a wide range of decision thresholds, even when the default threshold leads to missed positive cases.

Lab 6 – K-Nearest Neighbors (KNN)

Objective

This experiment aims to explore how the K-Nearest Neighbors (KNN) algorithm performs supervised classification based on proximity to labeled training data. The lab focuses on:

- Understanding the fundamental principles of instance-based (lazy) learning.
- Applying KNN to a synthetic multi-class dataset.
- Examining the importance of feature scaling in distance-based algorithms.
- Evaluating classification performance using accuracy.

Theory

K-Nearest Neighbors (KNN) is a supervised learning algorithm used for classification and regression. Unlike parametric models, KNN does not learn an explicit model during training. Instead, it stores all labeled training instances and defers computation until prediction time, which is why it is often referred to as a lazy learning algorithm.

For classification, KNN operates as follows:

- Given a new data point, compute the distance between this point and all training samples.
- Identify the k nearest neighbors based on a distance metric (commonly Euclidean distance).
- Assign the class label that appears most frequently among the k neighbors (majority voting).

The algorithm relies on the assumption that similar data points tend to belong to the same class. Because distance plays a central role, feature scaling is crucial to prevent attributes with larger numerical ranges from dominating the distance computation.

Code

The KNN classification code used in this experiment is available at the following GitHub repository: [KNN_Code](#)

Implementation on Synthetic Data

A synthetic two-dimensional dataset was generated using the `make_classification()` function from the scikit-learn library. The dataset consisted of 300 samples with two informative features and three distinct class labels.

The generated data was visualized using a scatter plot, where each color represented a different class. The plot revealed three reasonably well-separated groups, making the dataset suitable for multi-class classification using KNN.

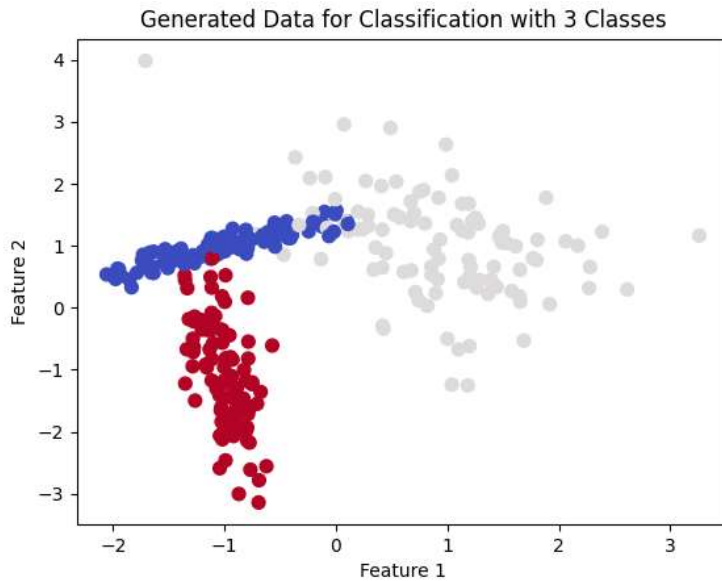
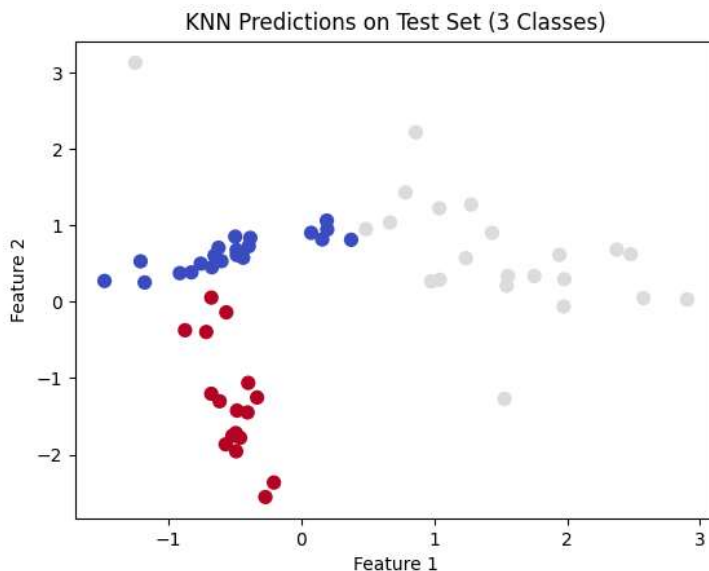


Figure 60. *Synthetic Data with 3 classes*

The dataset was then split into training (80%) and test (20%) sets. Feature standardization was applied using StandardScaler to ensure that both features contributed equally to distance calculations.

A KNN classifier with $k = 5$ was trained on the standardized training data and evaluated on the test set. The resulting classification accuracy was approximately 0.93, indicating that the algorithm was able to correctly classify most test samples.



```
"D:\Python\Orleans - AI Class\.venv\Scripts\python.exe" "D:\Python\Orleans - AI Class\6. knn.py"
Accuracy of KNN model with 3 classes: 0.93
```

Figure 61. *KNN classification results on the synthetic test dataset ($k = 5$).*

Effect of Feature Scaling

Because KNN relies on distance calculations, feature scaling has a significant impact on its performance. Without normalization, features with larger numeric ranges would dominate the distance metric and bias the classification results. By applying standardization, each feature was transformed to have zero mean and unit variance, resulting in balanced distance contributions and improved classification stability.

To analyze the impact of the parameter k , multiple values were tested:

$$k = \{1, 3, 5, 7, 9\}$$

For each value of k , the classifier was trained and evaluated on the test set. The accuracy values were recorded and plotted as a function of k .

The results show that:

- $k = 1$ produced lower accuracy due to sensitivity to noise and outliers.
- Increasing k improved stability by smoothing decision boundaries.
- Accuracy stabilized for $k \geq 3$, indicating a good balance between bias and variance.

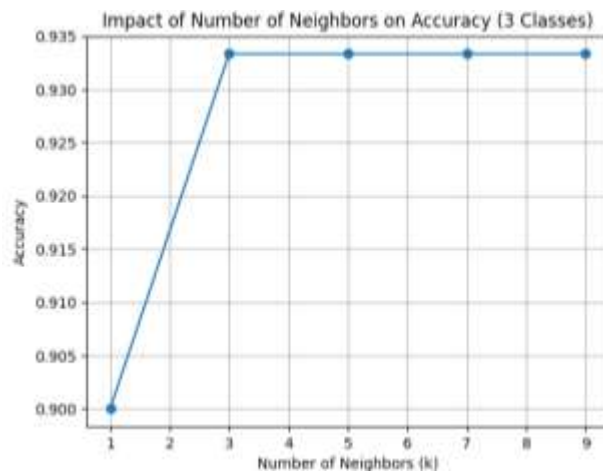


Figure 62. Classification accuracy as a function of the number of neighbors (k).

Conclusion

This experiment demonstrated the effectiveness of the K-Nearest Neighbors algorithm for supervised multi-class classification. Using a synthetic dataset, KNN achieved high classification accuracy when appropriate feature scaling and parameter tuning were applied.

The results highlighted the importance of selecting a suitable number of neighbors and standardizing features to ensure meaningful distance calculations. While KNN is simple and powerful for small to medium-sized datasets, its computational cost and sensitivity to dimensionality limit its scalability.

Lab 7 – Tree-Based Clustering (Silhouette-Based Splitting)

Objective

This experiment aims to ...